
Numbers

Natural numbers, principle of induction and complex numbers

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Introduction

In this module we will talk, as the name suggests, about numbers. In particular, we will put special emphasis on two specific sets of numbers: the natural and the complex numbers.

The properties of the set of natural numbers are basic to the most essential mathematical concepts, and their study is as old as human history.

In this module, we will introduce a technique for proving properties related to the natural numbers, the principle of induction. Proving that a certain property holds for infinite natural numbers is impossible if one tries to prove it one by one. Mathematical induction is a simple technique that allows us to ensure that a property is satisfied for all numbers, simply by seeing that it is satisfied for the first one and that the fact that it is satisfied for one makes the property verified for the next one. In the module, we will perform exercises to prove properties using this technique.

Regarding complex numbers, they are widely used in the mathematical modeling of physical processes. Among these processes, we can highlight the analysis of electric currents and electronic signals.

In this module, we will see what complex numbers are, how they are represented and how they are manipulated.

Objectives

The basic objectives that the student will achieve after working through the contents of this module are the following:

1. Understand the concept of proof by induction and know how to apply the principle of induction to prove properties of natural numbers.
2. Verify, using the principle of induction, that an algorithm performs the proposed task.
3. Know how to manipulate algebraic and geometric mathematical language, and understand its role in the expression of scientific and technological knowledge.
4. Know the set of complex numbers, know how to manipulate them and understand how they are applied in engineering.
5. To know how to analyze a situation and develop the ability to solve problems with the appropriate techniques.
6. Know basic mathematical software and know how to use it to perform calculations with different types of numbers.

1. The origin of numbers

Numbers, such as 2, -3 , the fraction $\frac{1}{2}$ or π , are human creations that have responded to different needs. For example, since ancient times, human beings have had the need to count:

- the number of sheep in a flock,
- the number of trees in a small plantation,
- the number of sons and daughters.

Thus arises the concept of the **natural numbers**, which we represent with the symbol $\mathbb{N}$, and that is a set formed by the numbers $1, 2, 3, \dots$. We denote this in this way:

$$\mathbb{N} = \{1, 2, 3, \dots\}$$

Occasionally, the number *zero* is also considered to be part of the set of natural numbers. In this case, we write

$$\mathbb{N}^* = \{0, 1, 2, 3, \dots\}$$

The number zero

Recall that, for example, the Romans had no symbol to represent the number *zero*.

To indicate that the number 3 is a natural number, we will write it as $3 \in \mathbb{N}$, which reads: “3 belongs to the set of natural numbers.”.

Since counting is the first need that all cultures, even the most ancient, have tried to cover, the first numbers handled are natural numbers.

Clearly, the natural number 2 can represent the fact “I have two sheep in my flock”. Now, if as a result of a commercial agreement, I go from having two sheep to owing two, how do we represent this fact? In this case, then, we can resort to the **integer numbers**, which are represented with the symbol $\mathbb{Z}$, and which are formed by the natural numbers, the natural numbers with negative sign and the zero. This set can be represented as follows:

$$\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}. \quad (1)$$

The next numerical set corresponds to that of the **rational numbers**, which is formed by the fractions and is represented by the symbol $\mathbb{Q}$. Its creation is

based on the need to be able to represent the parts of a totality, for example the fact of dividing an agricultural land in three equal parts as part of an inheritance. In this sense, we can represent this set in this way:

$$\mathbb{Q} = \left\{ \frac{a}{b} \text{ such that } a, b \in \mathbb{Z}, b \neq 0 \right\},$$

Remember that $a \in A$ is read “ a belongs to A ”.

that is, it is the set formed by the quotients or fractions of integers, where b is a non-zero integer.

Examples of rational numbers

Rational numbers are the numbers $\frac{1}{2}$ or $\frac{2}{-3}$, that is, numbers whose decimal expression has a repeating pattern. For example:

$$\frac{1}{2} = 0,5000 \dots = 0,5$$

$$\frac{2}{-3} = -0,6666 \dots = -0,\bar{6} \text{ or}$$

$$\frac{211}{990} = 0,21313 \dots = 0,2\overline{13}$$

But we can also write:

$$1 = \frac{1}{1} = \frac{2}{2}$$

$$-3 = \frac{-3}{1} = \frac{3}{-1}$$

i.e., also natural numbers and integers are rational numbers.

In this sense, we can write, $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q}$, which reads: “the set of natural numbers is a subset of the set of integers which, at the same time, is a subset of the set of rational numbers”. That is to say, every natural number is also an integer and every integer is, at the same time, a rational number.

Recall that $A \subset B$ is read “The set A is included in the set B ”.

Example

The equation $x + 2 = 0$ can represent a problem, such as: “What floor of a building am I on (x) if when I go up two floors (+2) I will be on the first floor (0)?”

The human need for numerical sets can also be stated in terms of equations. For example, the equation $x + 2 = 0$ can only be solved if we have the set of integers $\mathbb{Z}$, since the solution is the integer -2 , i.e., $x = -2 \in \mathbb{Z}$. In general, the equation

$$x + a = 0, \text{ where } a \text{ is a natural number}$$

has as its solution the integer $-a$, that is, $x = -a \in \mathbb{Z}$.

In the same way, the equation $2x = 3$ can only be solved thanks to the set of rational numbers $\mathbb{Q}$, since its solution is the rational number $\frac{3}{2}$, that is $x = \frac{3}{2} \in \mathbb{Q}$. In general, the equation

$$bx = a, \text{ where } a, b \text{ are integers and } b \neq 0$$

has as its solution the rational number $\frac{a}{b}$, i.e., $x = \frac{a}{b} \in \mathbb{Q}$.

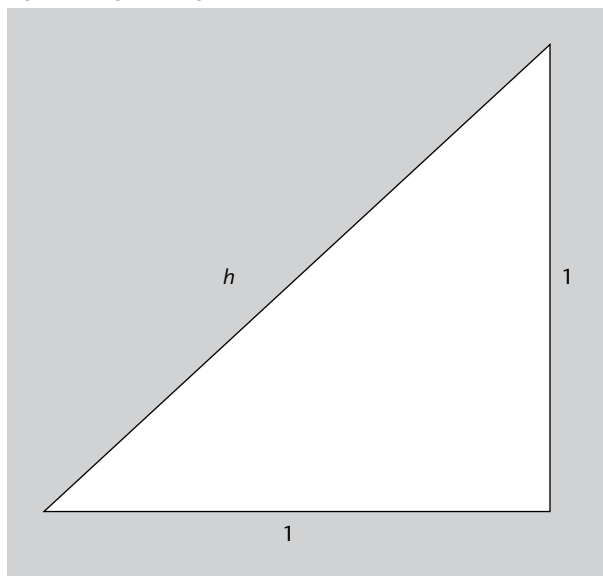
Example

The equation $2x = 3$ can answer, for example, the following problem: "Today I had two coffees ($2x$) in the bar of my town and I paid a total of 3 euros; what is the price of a coffee?".

1.1 Are there also equations that cannot be solved in the set of rational numbers $\mathbb{Q}$?

Imagine that we want to calculate the hypotenuse of a right triangle with legs of length 1, as shown in figure 1.

Figure 1. Right triangle



According to the Pythagorean theorem, the square of the hypotenuse is equal to the sum of the squares of the legs, that is:

$$h^2 = 1^2 + 1^2 = 2$$

The Greeks were already able to prove that the equation:

$$h^2 = 2$$

as simple as it may seem, it has no solution that can be expressed as a fraction. Now, we all know that one of the solutions of this equation is the number $\sqrt{2}$.

Pythagoras of Samos (ca. 582 B.C. – 507 B.C.), Greek mathematician and philosopher.

The other solution of $h^2 = 1^2 + 1^2 = 2$ is $-\sqrt{2}$.

It can be proved that the number $\sqrt{2}$ is not a rational number. Numbers that are not rational are called **irrational numbers** and are usually represented by the symbol $\mathbb{I}$.

This type of numbers represents a milestone in the history of geometry and were the beginning of a profound crisis of the Pythagorean sect in which they appeared. Indeed, the square, which, like the right triangle we have just seen, is one of the simplest geometrical figures, provides an awkward geometrical entity, the diagonal, which is not commensurable with the side. Thus, the possibility of always measuring exactly (with square and bevel!) was eliminated from geometry. The incommensurable numbers of the Greeks correspond to what we call irrational numbers.

Incommensurable diagonal

That the diagonal is incommensurable with the side means that it cannot be expressed as a fraction of the side.

Other examples of irrational numbers are π (approximately 3,1416), e (approximately 2,7183) or the square roots of the prime numbers $\sqrt{2}$, $\sqrt{3}$,...

Expression with decimal places

Remember that rational numbers, expressed in decimal form, are numbers with a finite number of decimals or with an infinite number of decimals but with a periodic pattern. For example:

$$\frac{3}{4} = 0,75, \quad \frac{5}{33} = 0,1515 \dots = 0,\overline{15}$$

On the contrary, irrational numbers, expressed in decimal form, have an infinite number of decimals without any periodicity. For example, the number π has infinite decimals without any periodicity:

$$\pi = 3,1415926535897932384626433832795028841971693993751 \dots$$

We define the set of the **real numbers**, that we represent with the symbol $\mathbb{R}$, as the union of the sets of rational numbers and irrational numbers, that is:

$$\mathbb{R} = \mathbb{Q} \cup \mathbb{I}$$

Remember that we use the symbol $\cup$ to designate the union of sets.

This expression reads “the set of real numbers is the union of the set of rational numbers and the set of irrational numbers”. Finally we have that:

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$$

It is interesting to note that, although we have called $\mathbb{N}$ the set of natural numbers, the real numbers $\sqrt{2}$ or π are also “natural”, in the sense that they are not a mathematical invention, but appear in nature. For example:

- What is the length of the diagonal of a square of side equal to one meter?
- What is the area of a circumference of radius equal to one meter?

1.2 Does every second-degree equation have a solution?

We have seen that to solve linear equations of the form $ax + b = 0$, in which a and b are integers, integers are not enough, but we must introduce the rational numbers. Likewise, we have seen that to solve some second-degree equations, such as $x^2 = 2$, we need the real numbers (which include the rational numbers).

The problem is now when we want to solve the equation:

$$x^2 = -1.$$

We know that there is no real number such that its square is a negative number. Therefore, it could be said that $x^2 = -1$ has no solutions.

Recall that to solve the equation $x^2 = 2$ we imagine the number $\sqrt{2}$; thus, analogously, to solve the equation $x^2 = -1$ we create a new number: $\sqrt{-1}$. This number is assigned the symbol i . Thus, to obtain solutions of the equation $x^2 = -1$ and of all the second-degree equations, $ax^2 + bx + c = 0$, we will need to create a new set: the set of **complex numbers**.

It is also common to use the symbol j to represent the number $\sqrt{-1}$.

What we propose is an important creative leap, since it is a matter of creating from “nothing” a solution for a type of equations, without being satisfied with the existence of equations without solution. The advantages of this creative leap outweigh the initial inconveniences.

2. Natural numbers: the principle of induction

As we have seen in the previous section, we could say that natural numbers have been invented to count, although they have other uses. For example, they are also used for sorting and are often used as identifier codes (ID number, telephone number, social security number, etc.). In engineering, it is often necessary to work with large masses of data: for example, think of a program that alphabetizes the census of a city. It is usually of interest to know that the program works correctly with any number of input data, and this is usually expressed by saying that a certain property $P(n)$ is verified (or holds) for any natural number n .

2.1 Introductory example

The following algorithm, which we will call the **MAX algorithm**, determines the highest number in a list:

Algorithm

A set of rules for solving a problem in a finite number of steps.

- **Input:** a list with n numbers $L = \{x_1, x_2, \dots, x_n\}$.
- **Output:** the highest number in the list L , which we will represent as $MAX(L)$, and which will be stored in the variable m .
- **Algorithm steps:**
 - 1) We take the first element of the list, x_1 , and initialize the variable m with this value. That is:

$$m = x_1$$

2) While we have not yet reached the last item on the list:

- we take the following element from the list, x_j
- if $x_j > m$, then $m = x_j$

Let us look at an example of the MAX algorithm application with $L = \{2, 17, 5, 3\}$.

- 1) We take the first element of the list, $x_1 = 2$, and we initialize m , $m = 2$.
- 2) While we have not yet reached the last element of the list, we take the next element and see if we have to update m :
 - We take the next element of the list, $x_2 = 17$. Since $17 > m$, since $m = 2$, we make $m = 17$.
 - We take the next item in the list, $x_3 = 5$. Since $5 < m$ (because m is now 17.), we do not need to do anything.

- We take the next element of the list, $x_4 = 3$. Since $3 < m$, we do not need to do anything, and we are done because we have reached the end of the list.

In this case, as the list is small, we can clearly see that the algorithm has worked correctly, but if we code it and use it to search the maximum of a large list (of millions of elements, for example), can we be certain that it will act correctly? The idea that allows us to reason the correctness of the algorithm is summarized in these points:

- It is evident that if $n = 1$, the algorithm calculates the maximum correctly.
- If for $n > 1$ the algorithm can correctly find the maximum m of a list of size n , as what it does next is to compare this number m with the new element of the list, it is clear that the result will be the maximum correctly found of the list of size $n + 1$.

This idea alludes to a mathematical proof technique widely used to prove properties of natural numbers: the principle of induction.

The **principle of induction** can be used when one wants to prove that a certain property is true for any natural number n and the technique consists of the following:

- Prove first that the property is true for $n = 1$.
- Then prove that if it is true for n , then it is true for $n + 1$.



Illustration of the principle of induction: if we push the first tile, as each tile will push the next one, we will know that all of them will fall.

It is as if we had a long row of dominoes tiles placed one after the other. If we push the first one and prove that each one as it falls pushes the next one, we will know that they will all fall. The principle of induction is based on the property of numbers being a well-ordered set.

2.2 The well-ordering principle (for natural numbers)

The set of natural numbers is a well-ordered set, which means that any subset of $\mathbb{N}$ has an element that is the smallest of all.

Example

The smallest element of the set $\mathbb{N}$ is 1.

The set of even numbers has one element, 2, which is the smallest element of all.

The smallest element of all the natural numbers that coincide with the sum of their divisors (except the number itself) is 6.

2.3 Basic formulation of the principle of induction

Let P be a property defined on the set of natural numbers that satisfies the following two conditions:

- 1) $P(1)$ holds.
- 2) For any n , if $P(n)$ holds, then $P(k + 1)$ also holds.

Then, the statement or property $P(n)$ holds for every natural number n .

Remark

The proof of the first property of the induction principle is called *base case*, and that of the second, *induction step*. It is often said that $P(n)$ is the induction hypothesis used to prove $P(n + 1)$.

Example

The principle of induction allows us to verify many conjectures concerning natural numbers. For example, how much is the sum of the first n odd numbers?

- If $n = 1$, the sum of the first odd number is 1.
- If $n = 2$, the sum of the first 2 odd numbers is $1 + 3 = 4$.
- If $n = 3$, the sum of the first 3 odd numbers is $1 + 3 + 5 = 9$.
- If $n = 4$, the sum of the first 4 odd numbers is $1 + 3 + 5 + 7 = 16$.

It appears that we always get n^2 , but is it true for any n ? We will formulate the property and prove it by induction.

Property $P(n)$: the sum of the n first odd numbers is n^2 .

We can formulate the property in algebraic terms to write it more precisely. To do this, we must realize that the n th odd number is $2n - 1$ (the first is 1, the second is 3, et cetera) and can be written: $P(n) : 1 + 3 + \dots + (2n - 1) = n^2$.

This is the result we will prove by induction:

1) Base case:

$P(1) : 1 = 1^2$, which obviously holds.

2) Induction step:

suppose that this induction hypothesis now holds,

$$P(n) : 1 + 3 + \dots + (2n - 1) = n^2$$

and from here we try to prove that:

$$P(n + 1) : 1 + 3 + \dots + (2n - 1) + (2(n + 1) - 1) = (n + 1)^2$$

To do this, we use the induction hypothesis and replace the n first summands of the first member by n^2 :

$$1 + 3 + \dots + (2n - 1) + (2n + 1) = n^2 + (2n + 1) = n^2 + 2n + 1 = (n + 1)^2$$

Therefore, we have proved that $1 + 3 + \dots + (2n - 1) = n^2$, and the proof is completed.

Problem 1

The story goes that when Gauss was about seven years old, his teacher, to keep them quiet for a while, made them add up all the natural numbers from 1 to 1000. That is to say, $1 + 2 + 3 + \dots + 1000$. In less than 5 minutes, Gauss had finished. He had added the first plus the last, the second, and the second to last, et cetera, and, seeing that $1 + 1000 = 2 + 999 = 3 + 998 = 1001$, he had simply made $500 \cdot 1001 = 500500$.

Prove by induction that Gauss's method is correct for any sum of n consecutive natural numbers.

The problem asks us to prove that the property $P(n) : 1 + 2 + \dots + n = \frac{n}{2}(n + 1)$ holds for any $n \in \mathbb{N}$.

Carl Friedrich Gauss
(1777-1855), German
mathematician, astronomer
and physicist.

1) Base case:

$P(1) : 1 = \frac{1(1+1)}{2} = \frac{2}{2} = 1$, that, obviously, holds.

2) Induction step:

Let us assume that this induction hypothesis holds: $P(n) : 1 + 2 + \dots + n = \frac{n}{2}(n+1)$ and from here we try to prove: $P(n+1) : 1 + 2 + \dots + (n+1) = \frac{(n+1)}{2}[(n+1)+1]$.

To do this, we use the induction hypothesis and substitute the n first summands of the first member by $\frac{n}{2}(n+1)$:

$$1 + 2 + \dots + (n+1) = \frac{n(n+1)}{2} + (n+1)$$

We use the common denominator to collect the expression on the right of the equality:

$$\begin{aligned} \frac{n(n+1)}{2} + (n+1) &= \frac{n(n+1)}{2} + \frac{2(n+1)}{2} = \\ \frac{n(n+1) + 2(n+1)}{2} &= \frac{(n+2)(n+1)}{2} = \frac{(n+1)}{2}[(n+1)+1] \end{aligned}$$

Then, we have proved that $1 + 2 + \dots + (n+1) = \frac{(n+1)}{2}[(n+1)+1]$. And we are done with the proof.

Remark

The expression $1 + 2 + \dots + n$ can be written in abbreviated form using the Σ notation as $\sum_{k=1}^n k$, that reads: "summation, from k equals 1 to n , of k ".

2.4 Application of the induction principle: verification of algorithms

In algorithmic analysis, induction is often used to prove the correctness of certain algorithms. For example, recall the MAX algorithm of the introductory example (subsection 2.1). Let us prove that it correctly performs the proposed task by reasoning by induction about the number n of data in the list:

1) Base case. It is obvious that if the list has only one data, MAX correctly calculates the maximum.

2) Induction step. Suppose the MAX algorithm correctly calculates the maximum of a list of size n and we consider a list of size $n+1$, $L = \{x_1, x_2, \dots, x_n, x_{n+1}\}$. To run the algorithm, the first thing to do is to remove the last element and use MAX to calculate the maximum of the list $\{x_1, x_2, \dots, x_n\}$. For the induction hypothesis, MAX correctly computes the maximum of this list, since it has size n . Let us consider $m = \text{MAX}\{x_1, x_2, \dots, x_n\}$. To finish, we only have to compare m with x_{n+1} and it is clear that the higher number of the two will be the maximum of L correctly calculated.

3. The set of complex numbers

3.1 Introductory example

Throughout history, several problems have appeared that had no solution in the set of real numbers. Here are some historical problems, in which the need for a new set of numbers was already beginning to be intuited: the set of complex numbers.

Suppose we want to solve this problem, which was posed by Diophantus of Alexandria in 275 BC.

Problem 2

Consider a right triangle as shown in the figure. Find the sides of the triangle (x, y, z) which make the perimeter equal to 12 and the area equal to 7.

This equation appears in the solution of this problem:

$$6x^2 - 43x + 84 = 0$$

where x is the length of one of the sides of the triangle.

The formula for solving a second-degree equation of the form $ax^2 + bx + c = 0$ is:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-b \pm \sqrt{\Delta}}{2a}$$

where the expression $b^2 - 4ac$ is called *discriminant*. In the case of the example we are considering, we can see that the discriminant is negative:

$$\Delta = b^2 - 4ac = (-43)^2 - 4 \cdot 6 \cdot 84 = 1849 - 2016 = -167 < 0$$

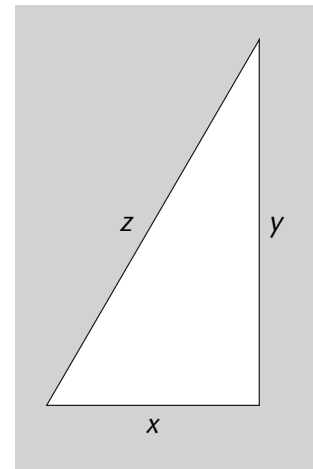
and, therefore, there are no real solutions. That is, we cannot construct any triangle of area 7 and perimeter 12.

Let us now consider this other problem, which was posed by Girolamo Cardano in 1545.

Problem 3

Divide the number 10 into two parts so that their product is 40. That is, we want to determine two numbers x and y such that their sum gives 10 and their product is 40.

The perimeter is the sum of the sides of any geometric figure. In this case, $x + y + z$.



Right triangle

In this case, the two equations that the problem poses are:

$$x + y = 10$$

$$xy = 40$$

in which, if we solve for y in the first equation and substitute it in the second equation, we obtain:

$$x(10 - x) = 40 \quad \text{or, equivalently,} \quad x^2 - 10x + 40 = 0$$

This equation also has no real solutions. But in this case, Cardano himself said that the two solutions were:

$$5 + \sqrt{-15} \quad \text{and} \quad 5 - \sqrt{-15}.$$

Gauss was the first to give the name **complex numbers** to this type of expressions.

As it has already happened to us before, now the set of real numbers is insufficient to find solutions to the quadratic equations. Therefore, we need to define the imaginary number.

We define the number i in such a way that $i^2 = -1$ or, in other words, $i = \sqrt{-1}$. The number i is called **imaginary number**.

Note now that, as the imaginary number i verifies that $i = \sqrt{-1}$, the solutions of the equation $x^2 - 10x + 40 = 0$ can be represented as

$$5 \pm \sqrt{-15} = 5 \pm \sqrt{15 \cdot (-1)} = 5 \pm \sqrt{15} \sqrt{-1} = 5 \pm \sqrt{15} i.$$

So, even if in the solution of a second-degree equation the discriminant is negative, we can always find the solutions using the imaginary number i . Remember that the number i allows us to work with roots of negative numbers.

Consequently, if we allow the solutions to be complex numbers, that is, numbers of the form:

$$a + bi$$

all the second-degree equations will have a solution.

René Descartes was the first to use the term *imaginary number* in 1637.

We will define the set of **complex numbers**, represented by the symbol $\mathbb{C}$, as the set formed by the expressions of the form $a + bi$, where a and b are real numbers and i is such that its square is equal to -1 . Formally:

$$\mathbb{C} = \{a + bi \text{ such that } a, b \text{ are real numbers}\}.$$

Complex numbers according to John Wallis

As a curiosity, John Wallis, English mathematician (1616-1703) said regarding complex numbers: "These imaginary quantities (as they are commonly called) arising from the *supposed* root of a negative square (when they happen) are reputed to imply that the case proposed is impossible." What do you think?

3.2 Representation of complex numbers: standard form

We have already seen that a complex number is a number having the form $a + bi$, where a and b are real numbers.

Examples of complex numbers

$$z_1 = 2 + 3i$$

$$z_2 = 3 - 4i$$

$$z_3 = -\frac{2}{3} + 8i$$

$$z_4 = \sqrt{2} + 10i$$

From the previous expressions, we can observe that a complex number is formed by the sum of two clearly differentiated parts that we define as **real part** (a) and **imaginary part** (b).

$$\underbrace{a}_{\substack{\text{real} \\ \text{part}}} + \underbrace{b}_{\substack{\text{imaginary} \\ \text{part}}} i,$$

Real and imaginary part representations

In some books you will see that the real part of $z = a + bi$ is also represented as $\text{Re}(z) = a$ and the imaginary part as $\text{Im}(z) = b$.

The following way of representing a complex number is called **standard form**:

$$z = a + bi$$

In table 1, you can find the real part and the imaginary part of the four complex numbers we have considered above. Notice that the imaginary number i is never part of either the real part or the imaginary part.

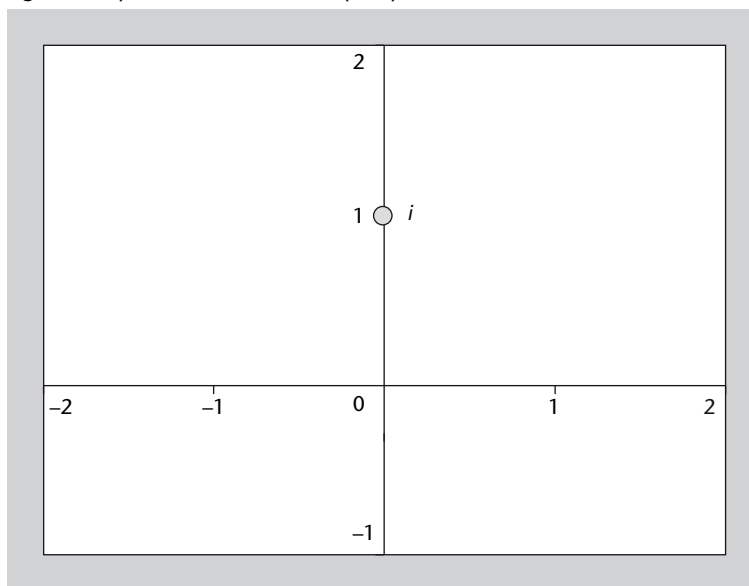
Table 1

Complex number	Real part	Imaginary part
$z_1 = 2 + 3i$	2	3
$z_2 = 3 - 4i$	3	-4
$z_3 = -2/3 + 8i$	-2/3	8
$z_4 = \sqrt{2} + 10i$	$\sqrt{2}$	10

The real numbers can be represented as points on a line (the real line). On the other hand, on the real line there is no place for $\sqrt{-1}$, that is, there is no place for i . How can we then represent the complex numbers graphically?

To solve this problem, Gauss wondered what would happen if he introduced a new direction (a vertical axis, like the y -axis) and if a point one unit away above the number line was used to locate i (see figure 2). All new numbers to solve any equation would now be combinations of i and usual numbers.

Figure 2. Representation of the complex plane

**Figure 2**

Representation of the complex plane. In particular, representation of the number i in the complex plane.

We have already seen that complex numbers consist of a real part and an imaginary part. In this way, as Gauss pointed out, each complex number can be represented by a single point in the plane, which is obtained in this way: the real part represents the **abscissa** (colloquially, x -coordinate) and the imaginary part is the **ordinate** (colloquially, y -coordinate). Look at figure 3. You can observe this correspondence in table 2.

Figure 3. Representation in the plane of the complex number $a + bi$.

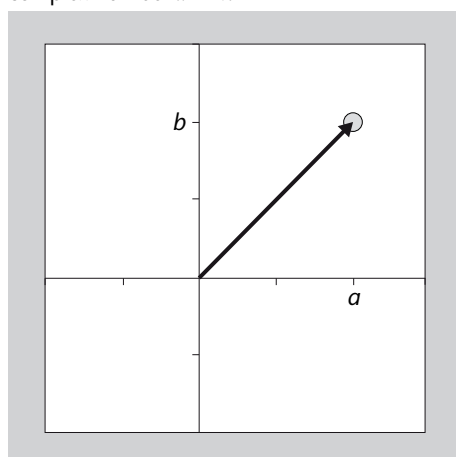


Table 2

Complex number	Real part	Imaginary part	Point in the plane
$z_1 = 2 + 3i$	2	3	(2, 3)
$z_2 = 3 - 4i$	3	-4	(3, -4)
$z_3 = -2/3 + 8i$	-2/3	8	(-2/3, 8)
$z_4 = \sqrt{2} + 10i$	$\sqrt{2}$	10	($\sqrt{2}$, 10)

Thus, the complex number $a+bi$ is identified with the point (a,b) in the plane.

- If a complex number has no real part (has real part equal to zero), it is called a **pure imaginary complex number**. For example, i , $2i$ or $\sqrt{3}i$ are pure imaginary complex numbers.
- If, on the other hand, a complex number has no imaginary part (has imaginary part equal to zero) it is called a **real number**. For example, 2, -1 or e are real numbers.

3.3 Operations with complex numbers

As in any set, we need to know what operations can be performed on its elements. In this case, for the set of complex numbers, we will see how we can add them, subtract them, multiply them, divide them, and so on.

3.3.1 Addition and subtraction of complex numbers in standard form

To add two complex numbers, we add the real parts and the imaginary parts separately.

Example (adding two complex numbers)

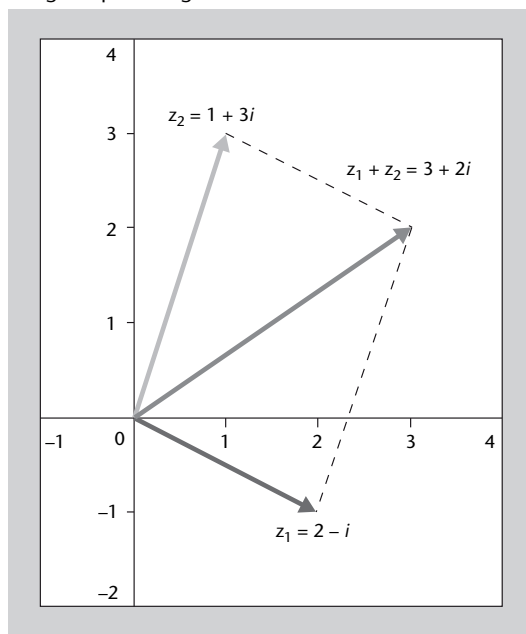
If we consider $z_1 = 1 + 2i$ and $z_2 = 4 + 3i$, then:

$$\begin{aligned} z_1 + z_2 &= (1 + 2i) + (4 + 3i) \\ &= (1 + 4) + (2 + 3)i \\ &= 5 + 5i. \end{aligned}$$

Notice that we treat the number i as if it were the variable of a polynomial, and we collect the terms that do not have i , on the one hand, and the terms that do have i , on the other.

In the complex plane, the numbers are added in the same way that we add the vectors of the plane. That is, the parallelogram law is used, as can be seen in figure 4, where the vectors $(2, -1)$ and $(1, 3)$ are added graphically.

Figure 4. Addition of two complex numbers using the parallelogram law



The subtraction of two complex numbers $z_1 - z_2$ where $z_1 = a + bi$ and $z_2 = c + di$ can be seen, at bottom, as a sum. In this way, the difference:

$$z_1 - z_2 = z_1 + (-z_2)$$

is nothing but the *sum* of the complex numbers z_1 and $-z_2$, where $-z_2 = -(c + di) = -c - di$.

Then, we can write:

$$\begin{aligned}
 z_1 - z_2 &= z_1 + (-z_2) \\
 &= (a + bi) + (-c - di) \\
 &= (a - c) + (b - d)i
 \end{aligned}$$

Example (subtracting two complex numbers)

If we consider $z_1 = 2 - i$ and $z_2 = 1 + 3i$, then

$$\begin{aligned}
 z_1 - z_2 &= z_1 + (-z_2) \\
 &= (2 - i) + (-1 - 3i) \\
 &= (2 - 1) + (-1 - 3)i \\
 &= 1 - 4i.
 \end{aligned}$$

The complex number $-z$ is called the **opposite** of the complex number z . From a geometrical point of view, the complex number $-z$ is the symmetric with respect to the origin of the complex number z , as can be seen in figure 5.

Figure 5. The complex number z and its opposite $-z$.

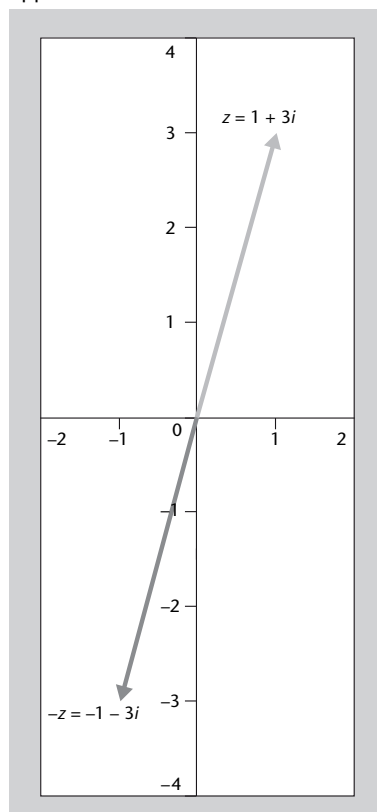
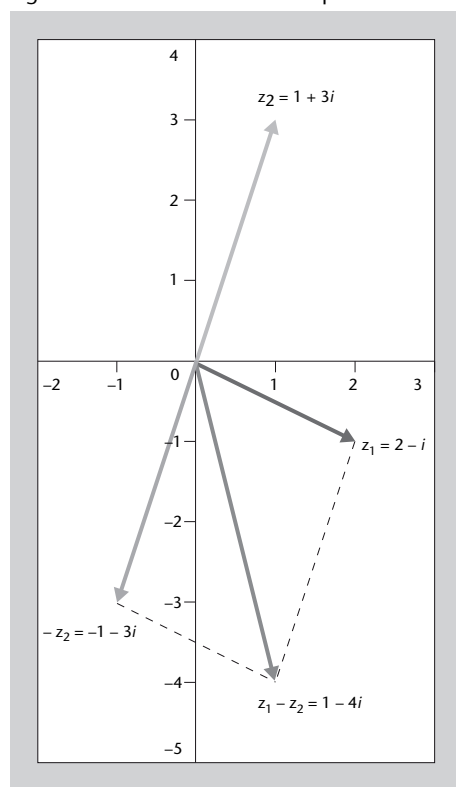


Figure 6. Subtraction of two complex numbers

**Figure 6**

To subtract two complex numbers $z_1 - z_2$, we consider the sum of the complex numbers $z_1 + (-z_2)$, where $-z_2$ is the opposite of the complex number z_2 .

3.3.2 Multiplying complex numbers in standard form

To multiply a complex number by a real number, we multiply the real and imaginary parts of the complex number by the real number.

Example (multiplying complex numbers)

If we consider $z = 1 + 2i$, to multiply it by the real number a , we do:

$$a \cdot z = a \cdot (1 + 2i) = a \cdot 1 + a \cdot 2i = a + 2ai$$

Recall that the real numbers are included in the complex numbers. In particular, every real number a can be written as $a + 0i$. Therefore, we have just seen how to multiply a complex number by another complex number with no imaginary part. Next, we will see how to multiply two complex numbers in which both can have non-zero imaginary part. But first, we will remember how to multiply the polynomials $p(x) = a + bx$ and $q(x) = c + dx$.

Two polynomials are multiplied by applying the distributive property, that is, multiplying everything by everything:

$$\begin{aligned} p(x) \cdot q(x) &= (a + bx) \cdot (c + dx) \\ &= ac + adx + bcx + bdx^2 \\ &= ac + (ad + bc)x + bdx^2 \end{aligned}$$

To multiply complex numbers we must proceed in the same way, considering that $i^2 = -1$.

Example

Let us consider $z_1 = 1 + 2i$ and $z_2 = 4 + 3i$:

$$\begin{aligned} z_1 \cdot z_2 &= (1 + 2i) \cdot (4 + 3i) \\ &= 1 \cdot (4 + 3i) + 2i \cdot (4 + 3i) \\ &= 1 \cdot 4 + 1 \cdot 3i + 2 \cdot 4i + 2 \cdot 3i^2 \\ &= 1 \cdot 4 + 1 \cdot 3i + 2 \cdot 4i - 2 \cdot 3 \quad (\text{where } i^2 = -1) \\ &= (1 \cdot 4 - 2 \cdot 3) + (1 \cdot 3 + 2 \cdot 4)i \\ &= -2 + 11i \end{aligned}$$

3.3.3 The conjugate of a complex number

The conjugate of a complex number, $z = a + bi$, is another complex number with the same real part but with the imaginary part equal in magnitude but opposite in sign. The conjugate of a complex number z is represented as $\bar{z}$.

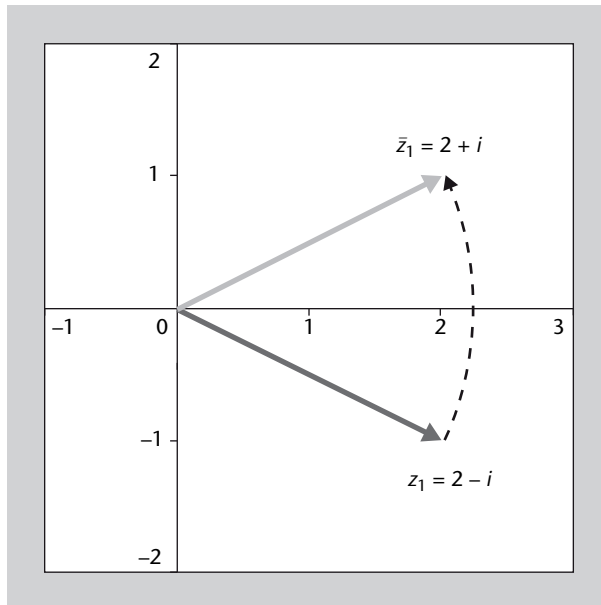
If we consider $z = 1 + 2i$, its conjugate is:

$$\bar{z} = 1 - 2i.$$

We will use the conjugate of a complex number mainly to divide complex numbers, as we will see later.

In the complex plane, the conjugate of a complex number is obtained by performing a symmetry with respect to the x -axis (figure 7).

Figure 7. The conjugate of a complex number

**Figure 7**

The conjugate of a complex number is obtained by performing a symmetry with respect to the x -axis.

Notice that, for any complex number, if we calculate the conjugate of the conjugate, we recover again the original complex number, i.e.:

$$z = 1 + 2i \longrightarrow \bar{z} = 1 - 2i \longrightarrow \overline{\bar{z}} = 1 + 2i = z.$$

The product of a complex number times its conjugate

Another important property is that if we multiply a complex number z by its conjugate $\bar{z}$, the result is a positive real number (or zero, when $z = 0$).

We will see later that this will be of great interest for dividing complex numbers.

Example

Let us consider $z = 1 + 2i$ and $\bar{z} = 1 - 2i$,

$$\begin{aligned} z \cdot \bar{z} &= (1 + 2i) \cdot (1 - 2i) \\ &= 1^2 - 1 \cdot 2i + 1 \cdot 2i - 2^2 i^2 \\ &= 1^2 - 1 \cdot 2i + 1 \cdot 2i + 2^2 \quad (\text{where } i^2 = -1) \\ &= 1^2 + 2^2 \geq 0 \end{aligned}$$

3.3.4 Dividing complex numbers in standard form

In the division of complex numbers, we will consider two cases:

- 1) The complex number that divides is a real number. For example:

$$\frac{z_1}{z_2} = \frac{1+2i}{5}$$

We note that this case is very simple to solve:

$$\frac{z_1}{z_2} = \frac{1+2i}{5} = \frac{1}{5} + \frac{2}{5}i$$

- 2) The number that divides is a complex number with non-zero imaginary part. For example:

$$\frac{z_1}{z_2} = \frac{1+2i}{3-4i}$$

In the more general case that the number that divides is a complex number with non-zero imaginary part, we will follow the following strategy: multiply and divide by the conjugate of the complex number of the denominator.

Thus, if we multiply both the numerator and the denominator by $\bar{z}_2$, the new denominator will be a real number. Therefore, it will be easy to perform the division:

$$\begin{aligned} \frac{z_1}{z_2} &= \frac{1+2i}{3-4i} \\ &= \frac{1+2i}{3-4i} \cdot \frac{3+4i}{3+4i} \\ &= \frac{(1+2i)(3+4i)}{(3-4i)(3+4i)} \\ &= \frac{3-8+6i+4i}{3^2+4^2} \\ &= \frac{-5+10i}{25} \\ &= -\frac{5}{25} + \frac{10}{25}i \\ &= -\frac{1}{5} + \frac{2}{5}i \end{aligned}$$

Remark

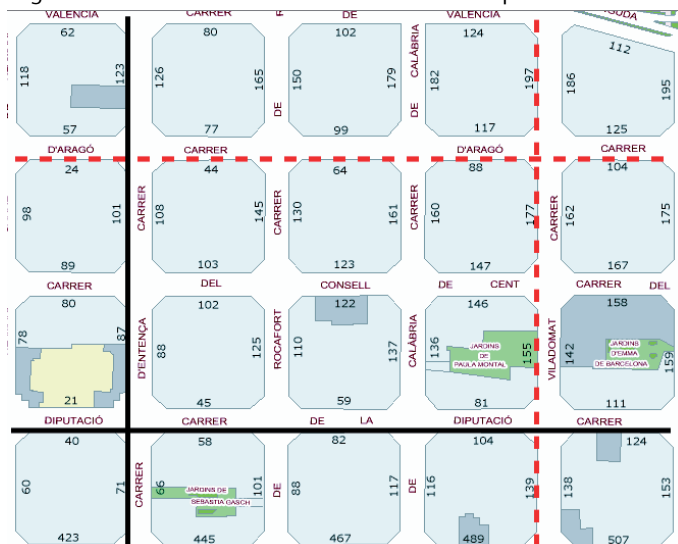
Multiplying and dividing by a value does not alter the final result.

3.4 Polar form: alternative way to represent complex numbers

A complex number, which we have so far represented in standard form $z = a + bi$ and which we have drawn as the point (a,b) in the plane, can also be represented in another way.

Cartesian coordinates or orthogonal coordinates, like those of the standard form, represent points by a pair of numbers: the x -coordinate, or abscissa (horizontal displacement), and the y -coordinate, or ordinate (vertical displacement). This type of coordinates can be very useful, for example, to indicate a location in the Eixample of Barcelona, as can be seen in figure 8. However, this is not the only way to represent points. As we have said, Cartesian coordinates can be useful for representing the surface of the Earth in a plane, but nevertheless ships use a two-dimensional radar system that places the points in the plane in circles centered on the origin, as can be seen in figure 9.

Figure 8. Cartesian coordinates in Barcelona's Eixample district

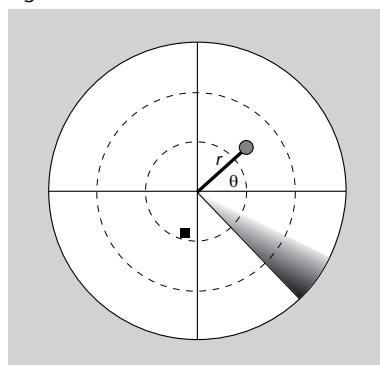


Source: Barcelona City Council, 2009

Figure 8

Considering the intersection between Diputació and Entença streets as the origin of coordinates, we will find the intersection between Aragó and Viladomat streets three streets to the right and two streets above, that is, at the point (3,2).

Figure 9. Polar coordinates on a radar



Thus, the representation of a point (a,b) in these new coordinates is based on (figure 10):

- The distance from the point (a,b) to the origin of coordinates, which is called the **magnitude** or **modulus**, and is usually represented as r .
- The angle formed by the straight line connecting the points (a,b) and $(0,0)$ with the positive x -axis (counterclockwise), which can also be called the **argument** and is represented as θ .

The following way of representing a complex number is called **polar form**:

$$r_{\theta}$$

Figure 10. From Cartesian coordinates to polar coordinates

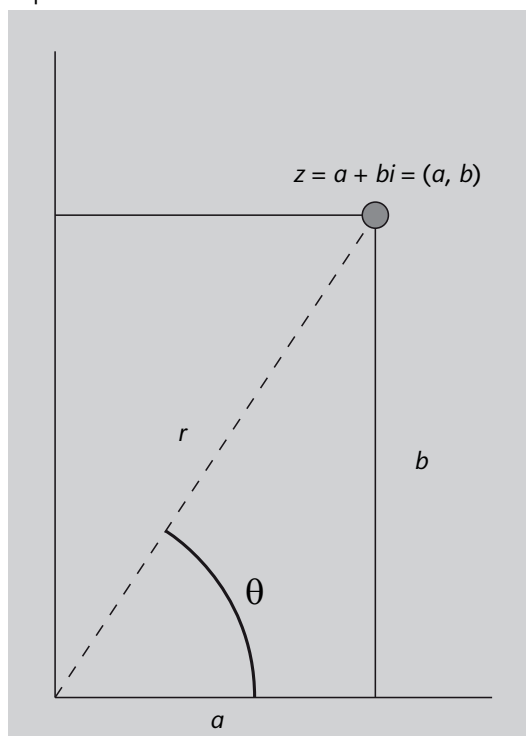


Figure 10

The complex number $a + bi$ can be expressed in polar form by the distance from the point to the origin, r , and the angle it forms with the positive x -axis, θ . That is: $a + bi = r_{\theta}$.

Example

Let us consider the complex numbers $3, 2i, -4$ and $-5i$. What is their representation in polar form?

- The complex number 3 has imaginary part equal to zero. This implies that it is located in the positive direction of the x -axis (in fact, it is a positive real number). Consequently, the angle formed by the straight line connecting the origin and this point with the positive x -axis is precisely 0. Moreover, the distance from this point to the origin is also 3. Therefore, we can write

$$3 = 3 + 0i = 3_0$$

- The complex number $2i$ has zero real part. Its representation as a point would be $(0,2)$, and this implies that it is located in the positive direction of the y -axis. On the other hand, the angle formed by the straight line connecting the origin and this point with the positive x -axis is a right angle, that is, it has a value of $\frac{\pi}{2}$ rad or 90° . Finally, the distance from this point to the origin is 2. Therefore, we can write:

$$2i = 0 + 2i = 2 \frac{\pi}{2} = 2_{90^\circ}$$

Remember that 360° is 2π radians.

- The complex number -4 has zero imaginary part and negative real part. This implies that it is located in the negative direction of the x -axis (in fact, it is a negative real number). Consequently, the angle formed by the straight line connecting the origin and this point with the positive x -axis is a straight angle, i.e., its value is π rad or 180° . On the other hand, the distance from this point to the origin is 4. Therefore, we can write:

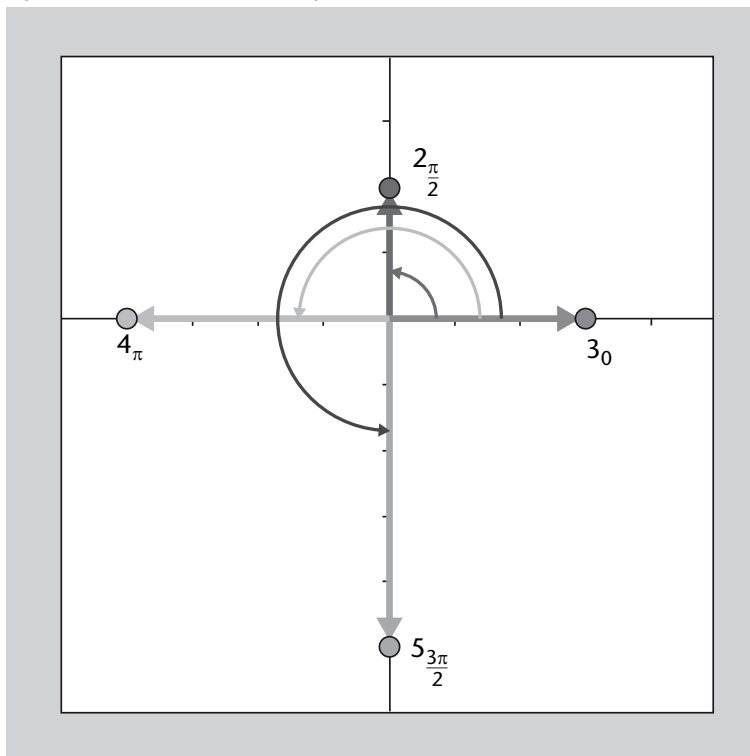
$$-4 = -4 + 0i = 4_\pi = 4_{180^\circ}$$

- The complex number $-5i$ has zero real part and negative imaginary part. Its representation as a point would be $(0, -5)$, and this implies that it is located in the negative direction of the y -axis. On the other hand, the angle that forms the straight line connecting the origin and this point with the positive x -axis is the sum of a right angle and a straight angle, that is to say, it has a value of $\frac{3\pi}{2}$ rad or 270° (if we measure the angles counterclockwise). The distance from this point to the origin is 5. Therefore, we can write:

$$-5i = 0 - 5i = 5 \frac{3\pi}{2} = 5_{270^\circ}$$

In this example, we have expressed in polar form the complex numbers that lie on the axes. In the next section, we will see how this can be generalized.

Figure 11. Polar form of the complex numbers $3, 2i, -4$ and $-5i$.



3.4.1 From standard to polar form

What is the relationship between the polar form and the standard form of a complex number? How can we go from one form to the other and vice versa?

Let us assume that we have the complex number in standard form $a + bi$, which we can represent graphically as the point (a,b) of the plane. Applying the Pythagorean theorem, we know that the distance from the point to the origin is:

$$r = \sqrt{a^2 + b^2}$$

In the same way, we know that the relation established between the angle formed by the straight line connecting the points (a,b) and $(0,0)$ with the positive x -axis and the same point (a,b) is determined by the expression:

$$\tan(\theta) = \frac{b}{a}$$

Therefore, θ can be obtained with the expression

$$\theta = \arctan\left(\frac{b}{a}\right), \quad \text{if } a \text{ (real part) is positive, or}$$

$$\theta = \arctan\left(\frac{b}{a}\right) + \pi, \quad \text{if } a \text{ is negative and } b \text{ (imaginary part) is positive, or}$$

$$\theta = \arctan\left(\frac{b}{a}\right) - \pi, \quad \text{if } a \text{ is negative and } b \text{ is also negative.}$$

In all cases, we will obtain an angle θ within the interval $[-\pi, \pi]$.

How do we calculate the arctangent?

To obtain the angle θ we can use a calculator or any symbolic calculation software, such as *CalcMe* or *Maple*.

If you calculate the arctangent with a calculator, you should know whether you use the `rad` mode (radians) or the `deg` mode (degrees).

Example (computing the arctangent)

Let us consider the following complex numbers:

$$z_1 = 1 + i$$

$$z_2 = -1 - i$$

The complex number z_1 lies in the *first quadrant*, since it has positive real part and positive imaginary part:

$$\operatorname{Re}(z_1) = 1 > 0, \quad \operatorname{Im}(z_1) = 1 > 0$$

On the other hand, the complex number z_2 lies in the *third quadrant*, since it has negative real part and negative imaginary part:

$$\operatorname{Re}(z_2) = -1 < 0, \quad \operatorname{Im}(z_2) = -1 < 0$$

The modulus of both complex numbers is:

$$r_1 = \sqrt{1^2 + 1^2} = \sqrt{2}, \quad r_2 = \sqrt{(-1)^2 + (-1)^2} = \sqrt{2}.$$

But what is their angle or argument?

- In the case of $z_1 = 1 + i$, since the real part is positive, the formula is:

$$\theta = \arctan\left(\frac{1}{1}\right) = \underbrace{\arctan(1)}_{\text{calculator}} = \frac{\pi}{4} \text{ rad}$$

- In the case of $z_2 = -1 - i$, since both the real part and the imaginary part are negative, the formula is:

$$\theta = \arctan\left(\frac{-1}{-1}\right) - \pi = \underbrace{\arctan(1)}_{\text{calculator}} - \pi = \frac{\pi}{4} - \pi = -\frac{3\pi}{4} \text{ rad}$$

Thus, the complex numbers $z_1 = 1 + i$, $z_2 = -1 - i$ can be represented in polar form as:

$$z_1 = 1 + i = \sqrt{2} \angle \frac{\pi}{4}$$

$$z_2 = -1 - i = \sqrt{2} \angle -\frac{3\pi}{4}$$

Polar representation is not unique

The polar form representation of a complex number is not unique. If we consider, for example, the complex number

$$z = 1 \angle \frac{\pi}{4},$$

and we add any multiple of 2π rad to the argument (i.e., we make complete turns), we get the same number. That is:

$$1 \angle \frac{\pi}{4} = 1 \angle \frac{\pi}{4} + 2\pi = 1 \angle \frac{\pi}{4} + 4\pi = 1 \angle \frac{\pi}{4} + 6\pi$$

In general terms, we can write (figure 12):

$$r_{\theta} = r_{\theta+2\pi k}, \quad k \in \mathbb{Z}$$

Figure 12. The polar form representation of a complex number is not unique.

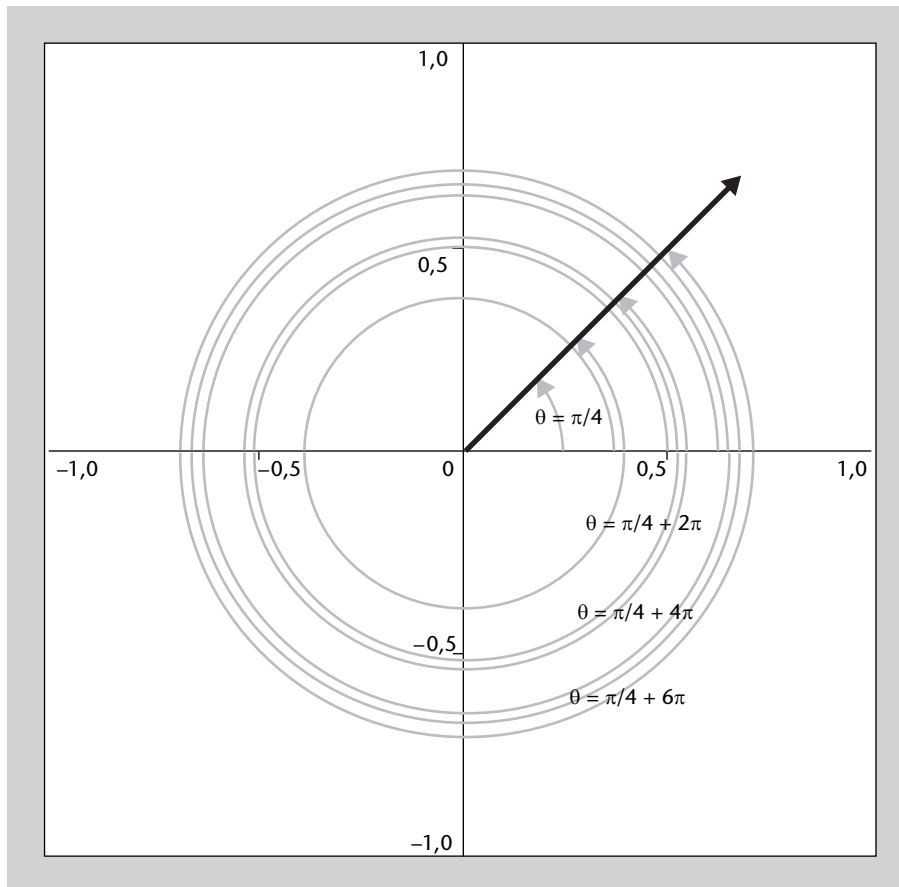


Figure 12

The polar representation of a complex number is not unique, since if we have a number r_{θ} and we add any multiple of 2π to the angle, we get the same number. In this figure we have represented $z = 1_{\pi/4} = 1_{\pi/4+2\pi} = 1_{\pi/4+4\pi} = 1_{\pi/4+6\pi}$

To unify the representation of complex numbers in polar form, it is usual to consider the angle θ that meets the following criteria

$$-\pi < \theta \leq \pi,$$

that is, of all the angles that can represent a complex number, we must choose the one that is between $-\pi$ and $+\pi$ radians. To do so, it may be necessary to subtract or add multiples of 2π radians.

When we want to transform a complex number from standard form to polar form, it is crucial to represent the complex number in the plane. In this way, we will know in which quadrant it is located, and we can avoid errors when calculating the argument.

3.4.2 From polar to standard form

Suppose we have a complex number $a + bi$ which is expressed according to its length and the angle it forms with the positive x -axis, i.e., $z = r_\theta$, as in figure 10.

Then, using the definitions of the trigonometric functions $\sin$ and $\cos$ we have:

$$\sin(\theta) = \frac{b}{r} \quad \text{y} \quad \cos(\theta) = \frac{a}{r}$$

Therefore:

$$a = r \cos(\theta), \quad b = r \sin(\theta)$$

That is, the number r_θ in standard form is:

$$z = r_\theta = \underbrace{r \cos(\theta)}_{\text{real part}} + \underbrace{r \sin(\theta)}_{\text{imaginary part}} i$$

The equivalence between the polar form and the standard form of a complex number is:

$$r_\theta = r \cos(\theta) + r \sin(\theta)i = r (\cos(\theta) + \sin(\theta)i)$$

The representation $r (\cos(\theta) + \sin(\theta)i)$ of a complex number is called the *trigonometric form*.

As a reminder, we show in table 3 the sine and cosine of the most common angles.

Table 3

θ	0	$\pi/6$	$\pi/4$	$\pi/3$	$\pi/2$
$\sin(\theta)$	0	$1/2$	$\sqrt{2}/2$	$\sqrt{3}/2$	1
$\cos(\theta)$	1	$\sqrt{3}/2$	$\sqrt{2}/2$	$1/2$	0

Example (from polar to standard form)

Let us consider the complex number in polar form $\sqrt{3} \frac{\pi}{6}$. What is its representation in standard form?

Directly applying the formula, we have that

$$\begin{aligned}
 \sqrt{3} \frac{\pi}{6} &= \sqrt{3} \cos\left(\frac{\pi}{6}\right) + \sqrt{3} \sin\left(\frac{\pi}{6}\right) i \\
 &= \sqrt{3} \frac{\sqrt{3}}{2} + \sqrt{3} \frac{1}{2} i \\
 &= \frac{3}{2} + \frac{\sqrt{3}}{2} i
 \end{aligned}$$

In the above calculation, we have looked up the cosine and sine values in table 3. For other angles, you can use a calculator or the instructions below.

For angles in the second, third and fourth quadrants, use these rules:

- If $\theta \in (\pi/2, \pi)$ (second quadrant), then

$$\begin{aligned}
 \sin(\theta) &= \sin(\pi - \theta) \\
 \cos(\theta) &= -\cos(\pi - \theta)
 \end{aligned}$$

- If $\theta \in (-\pi, -\pi/2)$ (third quadrant), then

$$\begin{aligned}
 \sin(\theta) &= -\sin(\pi + \theta) \\
 \cos(\theta) &= -\cos(\pi + \theta)
 \end{aligned}$$

- If $\theta \in (-\pi/2, 0)$ (fourth quadrant), then

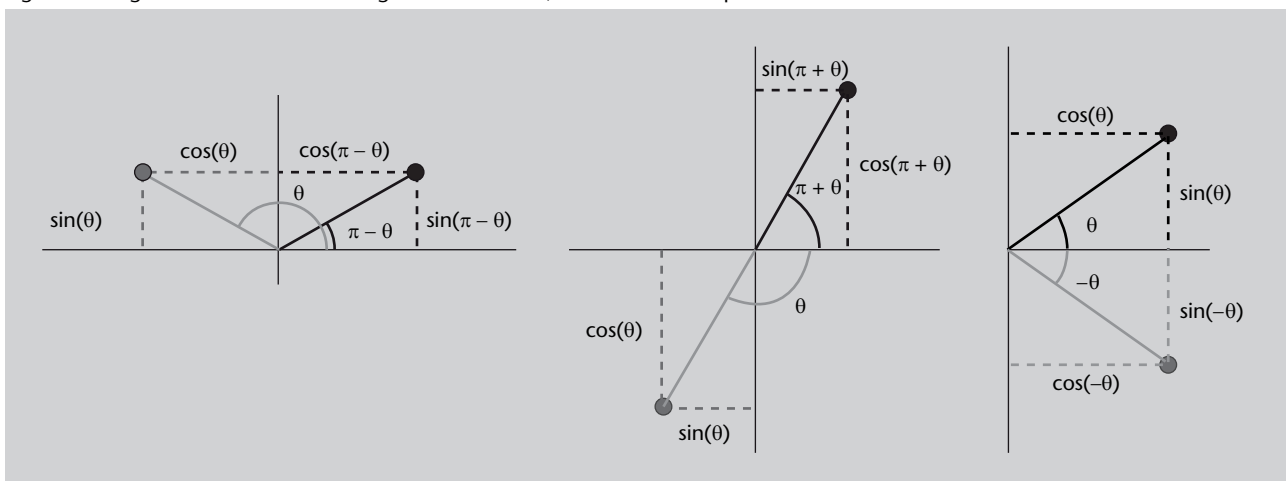
$$\begin{aligned}
 \sin(\theta) &= -\sin(-\theta) \\
 \cos(\theta) &= \cos(-\theta)
 \end{aligned}$$

Figure 13

Trigonometric relations to calculate the sine and cosine of angles in the second, third and fourth quadrants from the sines and cosines of angles in the first quadrant.

Figure 13 will help you to understand these rules.

Figure 13. Trigonometric relations of angles in the second, third and fourth quadrants



3.4.3 Arithmetic operations with complex numbers in polar form

Adding complex numbers in polar form

To add or subtract two complex numbers we must first express them in standard form, that is, write them in the form $z = a + bi$ and then add them as indicated in the previous subsection.

Thus, addition and subtraction operations are always performed in standard form. If desired, the final result can then be converted to polar form.

Example

Let us consider $z_1 = 2\pi$ and $z_2 = 5_{-\frac{\pi}{2}}$, two complex numbers in polar form, and we want to calculate:

$$z_1 + z_2, \quad \text{and} \quad z_1 - z_2$$

The first thing we will do is to convert both numbers to standard form:

$$\begin{aligned} z_1 &= 2 \underbrace{\cos(\pi)}_{-1} + 2 \underbrace{\sin(\pi)}_0 i = -2, \\ z_2 &= 5 \underbrace{\cos\left(-\frac{\pi}{2}\right)}_0 + 5 \underbrace{\sin\left(-\frac{\pi}{2}\right)}_{-1} i = -5i \end{aligned}$$

Then:

$$z_1 + z_2 = -2 + (-5i) = -2 - 5i$$

$$z_1 - z_2 = -2 - (-5i) = -2 + 5i$$

Multiplying and dividing complex numbers in polar form

To multiply two numbers in polar form, multiply the moduli and add the arguments.

Example

Let us consider $z_1 = 2\pi$ and $z_2 = 5_{-\frac{\pi}{2}}$, two complex numbers in polar form, and we want to calculate $z_1 \cdot z_2$.

Then:

$$\begin{aligned} z_1 \cdot z_2 &= 2_\pi \cdot 5_{-\frac{\pi}{2}} \\ &= (2 \cdot 5)_{\pi+(-\frac{\pi}{2})} \\ &= 10_{\frac{\pi}{2}} \end{aligned}$$

In a generic case, therefore, we can write:

$$r_{\theta_1} \cdot s_{\theta_2} = (r \cdot s)_{\theta_1+\theta_2}$$

Similarly, to divide two complex numbers in polar form, the moduli are divided and the arguments are subtracted.

Example

Let us consider $z_1 = 2_\pi$ and $z_2 = 5_{-\frac{\pi}{2}}$, two complex numbers in polar form, and we want to calculate $\frac{z_1}{z_2}$.

Then:

$$\begin{aligned} \frac{z_1}{z_2} &= \frac{2_\pi}{5_{-\frac{\pi}{2}}} \\ &= \left(\frac{2}{5}\right)_{\pi-(-\frac{\pi}{2})} \\ &= 0,4_{\frac{3\pi}{2}} \end{aligned}$$

In a generic case, therefore, we can write:

$$\frac{r_{\theta_1}}{s_{\theta_2}} = \left(\frac{r}{s}\right)_{\theta_1-\theta_2}$$

3.5 The exponential of a complex number

The exponential form of a complex number allows us to represent complex numbers in a very compact way. In addition, we can express the trigonometric functions sine and cosine in terms of complex numbers.

But let us start by recalling the two ways of representing complex numbers that we have seen so far:

- the standard form, $z = a + bi$, or
- the polar form, $z = r_{\theta} = r(\cos(\theta) + i\sin(\theta))$.

In this subsection, we will see a third way of representing complex numbers: the exponential form.

Let us consider the real exponential function, $\exp(x) = e^x$. This function can be rewritten as a sum of infinite terms:

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

Recall that the number e is defined as $e = \lim_{n \rightarrow \infty} (1 + \frac{1}{n})^n$

where $n! = n \cdot (n-1) \cdot (n-2) \cdots 2 \cdot 1$ is the factorial of the natural number n . Although this is a sum of infinite terms, in practice only a finite number of terms is usually used. For example, if we let us consider only six terms, and $x = 1$ we have that:

$$e^1 \approx 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \frac{1}{5!}$$

$$= 2 + 0,5 + 0,16667 + 0,04167 + 0,00833$$

$$= 2,71666$$

See also

This sum of infinite terms is called *power series expansion*.

The result is already good enough compared to the real value, which is $e = 2,71828$ (with five decimal places).

We can also express the trigonometric functions sine and cosine as an infinite sum of terms:

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

$$\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

We recover the expression of the exponential, e^x , but replacing the x by θi and finally rearranging the terms:

Remember that $i^2 = -1$.

$$\begin{aligned}
 e^{\theta i} &= 1 + (\theta i) + \frac{(\theta i)^2}{2!} + \frac{(\theta i)^3}{3!} + \frac{(\theta i)^4}{4!} + \frac{(\theta i)^5}{5!} + \dots \\
 &= 1 + \theta i - \frac{\theta^2}{2!} - \frac{\theta^3}{3!}i + \frac{\theta^4}{4!} + \frac{\theta^5}{5!}i + \dots \\
 &= \left[1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \dots \right] + \left[\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots \right] i \\
 &= \cos(\theta) + \sin(\theta)i
 \end{aligned}$$

Euler's Formula

The exponential of a pure imaginary complex number is therefore:

$$e^{\theta i} = \cos(\theta) + \sin(\theta)i$$

Since every complex number can be expressed as $z = r(\cos(\theta) + \sin(\theta)i)$ and, as we have just seen, $e^{\theta i} = \cos(\theta) + \sin(\theta)i$, then this allows us to represent the complex numbers in another way: $z = re^{\theta i}$, which we call the *exponential form*.

The exponential form of a complex number is:

$$z = re^{\theta i}$$

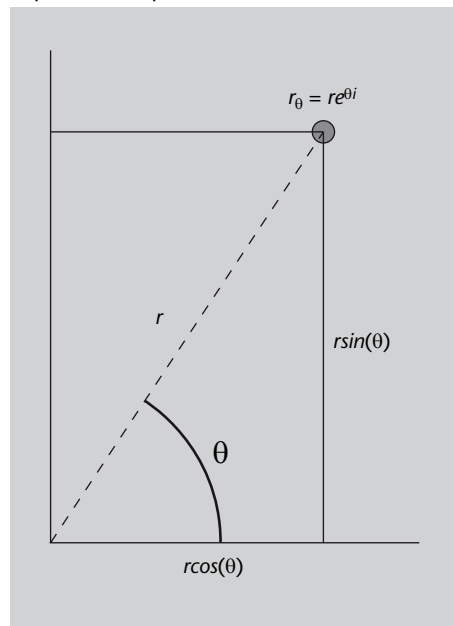
where $r = |z|$ is the modulus of z and θ is the argument of z .

The polar representation of a complex number is thus equivalent to the exponential representation of the same complex number. In fact:

$$re^{\theta i} = r(\cos(\theta) + i \sin(\theta)) = r_{\theta}$$

See figure 14.

Figure 14. Representation of a complex number in polar and exponential form



Example

Table 4 represents the following equivalences between complex numbers expressed in standard form:

$$z_1 = 1 + i, \quad z_2 = i, \quad z_3 = 1 + \sqrt{3}i, \quad z_4 = 1 - \sqrt{3}i$$

Table 4

Standard	Polar	Exponential
$z_1 = 1 + i$	$z_1 = \sqrt{2}e^{i\frac{\pi}{4}}$	$z_1 = \sqrt{2}e^{i\frac{\pi}{4}}$
$z_2 = i$	$z_2 = 1e^{i\frac{\pi}{2}}$	$z_2 = e^{i\frac{\pi}{2}}$
$z_3 = 1 + \sqrt{3}i$	$z_3 = 2e^{i\frac{\pi}{3}}$	$z_3 = 2e^{i\frac{\pi}{3}}$
$z_4 = 1 - \sqrt{3}i$	$z_4 = 2e^{-i\frac{\pi}{3}}$	$z_4 = 2e^{-i\frac{\pi}{3}}$

3.5.1 Operations of complex numbers in exponential form

The operations of complex numbers in exponential form are therefore equivalent to the operations we have already seen in polar form. These are:

- **Product.** The moduli are multiplied and the arguments are added:

$$\begin{aligned}
 2e^{i\frac{\pi}{3}} \cdot 3e^{i\frac{\pi}{2}} &= (2 \cdot 3)e^{i(\frac{\pi}{3} + \frac{\pi}{2})} \\
 &= 6e^{i\frac{5\pi}{6}}
 \end{aligned}$$

In the general case:

$$r_1 e^{\theta_1 i} \cdot r_2 e^{\theta_2 i} = r_1 r_2 e^{(\theta_1 + \theta_2) i}$$

- **Division.** The modules are divided, and the arguments are subtracted:

$$\begin{aligned} \frac{2e^{\frac{\pi}{3}i}}{3e^{\frac{\pi}{2}i}} &= \frac{2}{3} e^{(\frac{\pi}{3} - \frac{\pi}{2})i} \\ &= \frac{2}{3} e^{-\frac{\pi}{6}i} \end{aligned}$$

In the general case:

$$\frac{r_1 e^{\theta_1 i}}{r_2 e^{\theta_2 i}} = \frac{r_1}{r_2} e^{(\theta_1 - \theta_2) i}$$

- **Complex conjugate.** The conjugate of a complex number expressed in exponential form is another complex number with the same modulus but with the opposite argument:

$$\overline{2e^{\frac{\pi}{3}i}} = 2e^{-\frac{\pi}{3}i}$$

In the general case:

$$\overline{r e^{\theta i}} = r e^{-\theta i}$$

- **Addition and subtraction.** To add or subtract complex numbers in exponential form, we must express them in standard form:

$$\begin{aligned} r_1 e^{\theta_1 i} + r_2 e^{\theta_2 i} &= r_1 (\cos(\theta_1) + i \sin(\theta_1)) + r_2 (\cos(\theta_2) + i \sin(\theta_2)) \\ &= (r_1 \cos(\theta_1) + r_2 \cos(\theta_2)) + i (r_1 \sin(\theta_1) + r_2 \sin(\theta_2)) \end{aligned}$$

The introduction of the exponential form also allows us to calculate powers of complex numbers in a very simple way, using the properties of the exponential function that we already knew when we worked with real numbers:

- **Powers.** To *square* a complex number in exponential form we use the definition of the product of complex numbers in polar form:

$$\begin{aligned}\left(re^{\theta i}\right)^2 &= re^{\theta i} \cdot re^{\theta i} = r^2 e^{(\theta+\theta)i} \\ &= r^2 e^{2\theta i}\end{aligned}$$

In other words, what we have done is to multiply the moduli ($r \cdot r = r^2$) and add up the arguments ($\theta + \theta = 2\theta$). Thus, we have managed to have:

$$\left(re^{\theta i}\right)^2 = r^2 e^{2\theta i}$$

Likewise, to raise a complex number in exponential form to the power n , we raise to the power n the modulus and multiply by n the argument. In effect:

$$\begin{aligned}\left(re^{\theta i}\right)^n &= \underbrace{re^{\theta i} \cdot re^{\theta i} \cdots re^{\theta i}}_{n \text{ times}} = \underbrace{(r \cdots r)}_{n \text{ times}} e^{\overbrace{(\theta + \cdots + \theta)}^{n \text{ times}} i} \\ &= r^n e^{n\theta i}\end{aligned}$$

In other words, we have reached the expression:

$$\left(re^{\theta i}\right)^n = r^n e^{n\theta i}$$

3.6 Roots of unity

In the set of real numbers, a square root of a positive number has two solutions. For example, the two square roots of the number 1 are -1 and $+1$, since:

$$1^2 = 1$$

$$(-1)^2 = 1$$

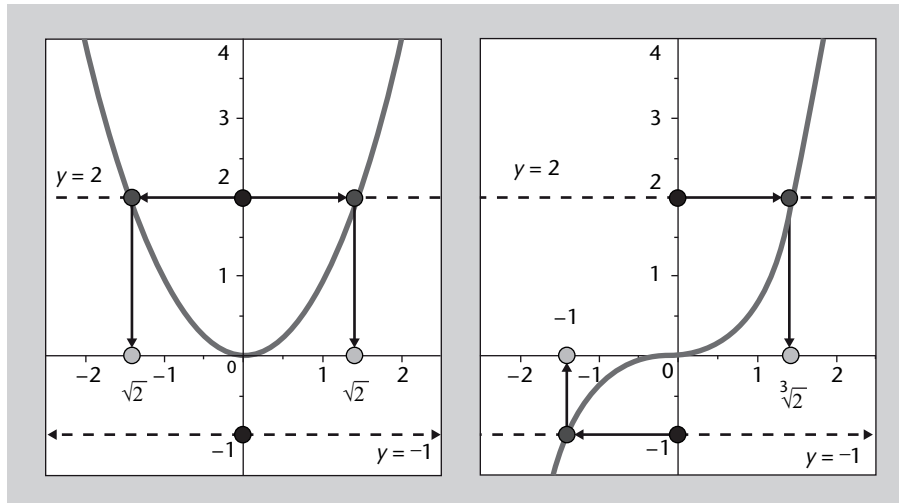
If we are trying to calculate a cube root, there is only one solution. For example, the cube root of -1 is -1 itself:

$$(-1)^3 = -1$$

This can be seen in figure 15:

- When we calculate square roots, for example of 2, what we do is to find which numbers squared equal 2. These numbers are, as you can see, $-\sqrt{2}$ and $\sqrt{2}$. We cannot calculate square roots of negative numbers.
- When we calculate cube roots, for example of 2, what we do is to look for which numbers raised to the cube equal 2. In this case, we have only one value, $\sqrt[3]{2}$ (approximately 1,26), as seen in figure 15. All numbers, both positive and negative, have a unique cube root.

Figure 15. Square and cube roots

**Figure 15**

Square roots of a real number (left): to calculate the square roots of a number s we look for the intersections of the parabola $y = x^2$ with the straight line $y = s$. Cubic roots of a real number (right): to determine the cubic roots of a number s we look for the intersections of the function $y = x^3$ with the line $y = s$.

In the set of real numbers:

- Roots of even index (square root, fourth root, etc.) have no solution if the number is negative.
- Roots of even index (square root, fourth root, etc.) have two solutions if the number is positive.
- Roots of odd index (cube root, fifth root, etc.) always have one solution.

In the set of complex numbers, there are differences with respect to what happens in the set of real numbers:

- A square root always has two solutions;
- A cube root always has three solutions;
- And, in general, an n th root always has n solutions.

3.6.1 Cube roots of unity

Let us consider, for example, the problem of finding the complex numbers z that satisfy the equation $z^3 = 1$. In this case, the solutions of this equation, that is, the numbers z that verify the above equation, are called **cubic roots of unity**. To solve the previous equation, the first thing we must do is to consider that the number 1 is a complex number that in polar form has modulus 1 and argument 0:

$$1 = 1_0 = 1e^{0i}$$

In addition, we can use other equivalent arguments, such as 2π , 4π , 6π or any other multiple of 2π , since by adding a complete turn we are not changing the position of the vector that represents this complex number and, therefore, we do not change its value either. In this way, we can also write:

$$\begin{aligned} 1 &= 1_0 = 1_{2\pi} = 1_{4\pi} = 1_{6\pi} = \dots \\ &= e^{0i} = e^{2\pi i} = e^{4\pi i} = e^{6\pi i} = \dots \\ &= e^{2k\pi i}, \quad k = 0, 1, 2, 3, \dots \end{aligned}$$

Thus, the equation $z^3 = 1$ is equivalent to:

$$z^3 = e^{2k\pi i}, \quad k = 0, 1, 2, 3, \dots$$

To solve this equation, what we do is to add cube roots to both sides of the equation or, what is the same, to raise both parts of the equality to the power $\frac{1}{3}$. If we do this, we get this equation:

$$\left(z^3\right)^{\frac{1}{3}} = z = e^{\frac{2k\pi i}{3}}, \quad k = 0, 1, 2, 3, \dots$$

Finally, to determine the solutions, we only need to assign values to the k . In this way, we will find the values of the different cube roots. In particular, since we know that we must find exactly 3 cube roots, we substitute the value of k for 0, 1 and 2:

$$k = 0 : \quad z = e^{0i} = 1$$

$$k = 1 : \quad z = e^{\frac{2\pi}{3}i} = \cos\left(\frac{2\pi}{3}\right) + i \sin\left(\frac{2\pi}{3}\right) = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$$

$$k = 2 : \quad z = e^{\frac{4\pi}{3}i} = \cos\left(\frac{4\pi}{3}\right) + i \sin\left(\frac{4\pi}{3}\right) = -\frac{1}{2} - \frac{\sqrt{3}}{2}i$$

In the same way that the cube roots of unity are calculated, the roots of any complex number can be calculated.

Equation $z^n = re^{\theta i}$ has n solutions, which are:

$$z = \sqrt[n]{r} e^{\left(\frac{\theta}{n} + \frac{2\pi}{n}k\right)i}, \quad k = 0, 1, 2, 3, \dots, n-1$$

Notice that, since r is a positive real number, $\sqrt[n]{r}$ refers to the positive n th root of r .

Example. Cube roots of -8

Let us consider now the problem of finding the complex numbers z that satisfy the equation:

$$z^3 = -8$$

In this case, the solutions of this equation, i.e., the numbers z that verify the above equation, are called the *cube roots* of -8 .

To solve the above equation, the first thing we must do is to keep in mind that the number -8 is a complex number that in polar form has modulus 8 and argument π :

$$-8 = 8\pi = 8e^{\pi i}$$

As before, we can use other equivalent arguments, e.g. $\pi + 2\pi$, $\pi + 4\pi$ or $\pi + 6\pi$, since we are adding complete turns. In this way, we can also write:

$$\begin{aligned} -8 &= 8\pi = 8_{\pi+2\pi} = 8_{\pi+4\pi} = 8_{\pi+6\pi} = \dots \\ &= 8e^{\pi i} = 8e^{(\pi+2\pi)i} = 8e^{(\pi+4\pi)i} = 8e^{(\pi+6\pi)i} = \dots \\ &= 8e^{(\pi+2k\pi)i}, \quad k = 0, 1, 2, 3, \dots \end{aligned}$$

Thus, the equation $z^3 = -8$ is equivalent to:

$$z^3 = 8e^{(\pi+2k\pi)i}, \quad k = 0, 1, 2, 3, \dots$$

To solve this equation, what we do is to add cube roots to both sides of the equation or, what is the same, to raise both parts of the equality to the power $\frac{1}{3}$. If we do this, we get this equation:

$$\left(z^3\right)^{\frac{1}{3}} = z = \underbrace{\sqrt[3]{8}}_2 e^{\left(\frac{\pi}{3} + \frac{2k\pi}{3}\right)i}, \quad k = 0, 1, 2, 3, \dots$$

Finally, to determine the solutions, we only need to assign values to the k . In this way, we will find the values of the different cube roots. In particular, since we know that we must find exactly 3 cube roots, we substitute the value of k for 0, 1 and 2:

$$k = 0 : \quad z = \sqrt[3]{8}e^{\frac{\pi}{3}i} = 2 \left(\cos \left(\frac{\pi}{3} \right) + i \sin \left(\frac{\pi}{3} \right) \right)$$

$$= 2 \left(\frac{1}{2} + \frac{\sqrt{3}}{2}i \right) = 1 + \sqrt{3}i$$

$$k = 1 : \quad z = \sqrt[3]{8}e^{\pi i} = 2 (\cos (\pi) + i \sin (\pi))$$

$$= 2(-1) = -2$$

$$k = 2 : \quad z = \sqrt[3]{8}e^{\frac{5\pi}{3}i} = 2 \left(\cos \left(\frac{5\pi}{3} \right) + i \sin \left(\frac{5\pi}{3} \right) \right)$$

$$= 2 \left(\frac{1}{2} - \frac{\sqrt{3}}{2}i \right) = 1 - \sqrt{3}i$$

Example. Square, cube and fourth roots of -1

We first express the complex number -1 in exponential form:

$$-1 = e^{\pi i} = e^{\pi i + 2\pi k i}, \quad k = 0, 1, 2, 3, \dots$$

Therefore, we want to solve the equation:

$$z^n = e^{\pi i + 2\pi k i} = e^{(\pi + 2\pi k)i}, \quad k = 0, 1, 2, 3, \dots$$

for $n = 2, 3$, and 4. To do this at the same time, we apply the n th root to both sides of the equality:

$$(z^n)^{\frac{1}{n}} = e^{\frac{(\pi + 2\pi k)i}{n}}$$

$$z = e^{\left(\frac{\pi}{n} + \frac{2\pi k}{n}\right)i}, \quad k = 0, 1, 2, 3, \dots$$

To determine the square roots, we substitute $n = 2$ and consider $k = 0$ and $k = 1$.

$$k = 0 : \quad z = e^{\frac{\pi}{2}i} = \cos \left(\frac{\pi}{2} \right) + i \sin \left(\frac{\pi}{2} \right) = i$$

$$k = 1 : \quad z = e^{\left(\frac{\pi}{2} + \pi\right)i} = e^{\frac{3\pi}{2}i} = \cos \left(\frac{3\pi}{2} \right) + i \sin \left(\frac{3\pi}{2} \right) = -i$$

Similarly, to determine the cube roots, we substitute $n = 3$ and consider $k = 0, k = 1$ and $k = 2$.

$$k = 0 : \quad z = e^{\frac{\pi}{3}i} = \cos\left(\frac{\pi}{3}\right) + i \sin\left(\frac{\pi}{3}\right) = \frac{1}{2} + \frac{\sqrt{3}}{2}i$$

$$k = 1 : \quad z = e^{\left(\frac{\pi}{3} + \frac{2\pi}{3}\right)i} = e^{\pi i} = -1$$

$$k = 2 : \quad z = e^{\left(\frac{\pi}{3} + \frac{4\pi}{3}\right)i} = e^{\frac{5\pi}{3}i} = \cos\left(\frac{5\pi}{3}\right) + i \sin\left(\frac{5\pi}{3}\right) = \frac{1}{2} - \frac{\sqrt{3}}{2}i$$

Finally, to determine the fourth roots, we substitute $n = 4$ and consider $k = 0, k = 1, k = 2$ and $k = 3$.

$$k = 0 : \quad z = e^{\frac{\pi}{4}i} = \cos\left(\frac{\pi}{4}\right) + i \sin\left(\frac{\pi}{4}\right) = \frac{1}{2} + \frac{\sqrt{2}}{2}i = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i$$

$$k = 1 : \quad z = e^{\left(\frac{\pi}{4} + \frac{2\pi}{4}\right)i} = e^{\frac{3\pi}{4}i} = \cos\left(\frac{3\pi}{4}\right) + i \sin\left(\frac{3\pi}{4}\right) = -\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i$$

$$k = 2 : \quad z = e^{\left(\frac{\pi}{4} + \frac{4\pi}{4}\right)i} = e^{\frac{5\pi}{4}i} = \cos\left(\frac{5\pi}{4}\right) + i \sin\left(\frac{5\pi}{4}\right) = -\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i$$

$$k = 3 : \quad z = e^{\left(\frac{\pi}{4} + \frac{6\pi}{4}\right)i} = e^{\frac{7\pi}{4}i} = \cos\left(\frac{7\pi}{4}\right) + i \sin\left(\frac{7\pi}{4}\right) = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i$$

In figure 16, we have plotted the square, cube and fourth roots of the number $z = -1$. Notice that all the roots have the same modulus (they are located on a circle) and that they are located at the vertices of a regular polygon of n sides. That is, the square roots are always aligned, the cube roots are the vertices of an equilateral triangle, and the fourth roots are the vertices of a square. In figure 17, we have represented the fifth and sixth roots of $z = -1$, which are also the vertices of a regular polygon.

Figure 16. Square, cubic and fourth roots of the number $z = -1$

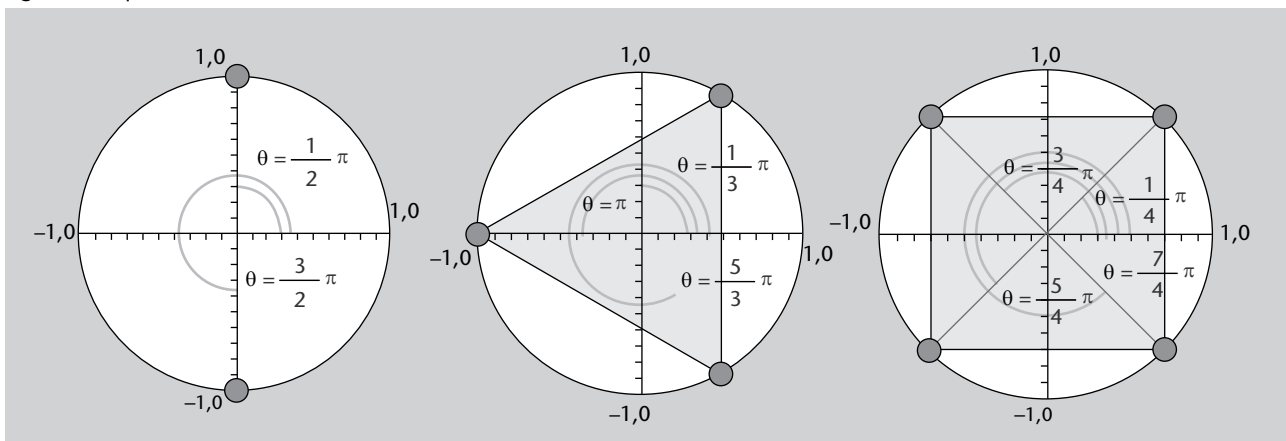
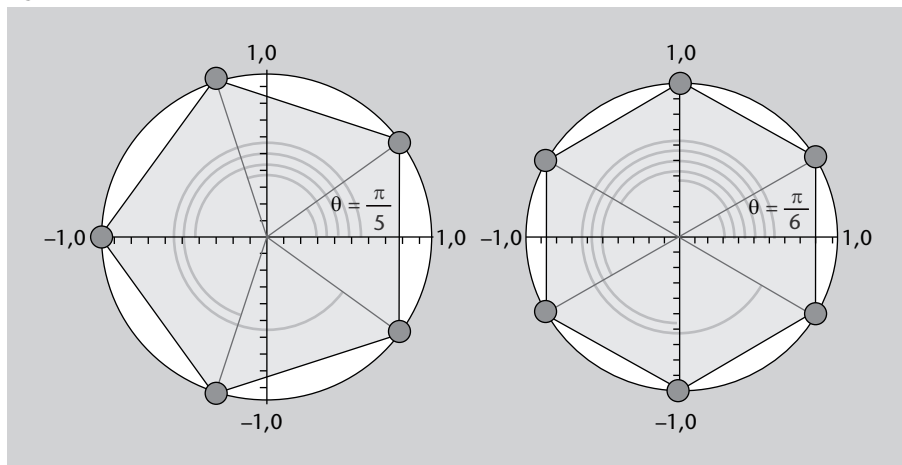


Figure 17. Fifth and sixth roots of the number $z = -1$ 

Summary

In this module, we have worked with two sets of numbers: natural numbers and complex numbers.

As for the natural numbers, we have seen that they are characterized by being an ordered set, and we have studied their principle of mathematical induction and also its application to the verification of algorithms.

The basic formulation of the principle of mathematical induction is the following: let P be a property defined on the set of natural numbers that satisfies the following two conditions:

- 1) $P(1)$ holds.
- 2) For any n , if $P(n)$ holds, then $P(n + 1)$ also holds.

Then, the property holds for every natural number.

As for complex numbers, we have seen that they cannot be represented on a line because they have two dimensions. These numbers are represented on a plane composed of a real axis and an imaginary axis. The unit of the imaginary axis is the number $i = \sqrt{-1}$. We have studied three possible types of representations of the complex numbers:

- 1) Standard form
- 2) Polar form
- 3) Exponential form

We have learned to perform operations with complex numbers in each of these representations: addition, subtraction, product, conjugation of a complex number and division.

Finally, we have seen how the different roots of unity can be calculated within the set of complex numbers (square roots, cube roots, fourth roots, etc.), and, analogously, we have also calculated different roots of other complex numbers.

Self-assessment exercises

1. Prove by induction that for every $n \in \mathbb{N}$ we have that $1^3 + 2^3 + \dots + n^3 = \frac{n^2(n+1)^2}{4}$.
2. Prove by induction that for every $n \in \mathbb{N}$ the number $n(n+1)$ is an even number.
3. Calculate $\sum_{k=1}^n \frac{1}{k(k+1)}$ for different values of $n \in \mathbb{N}$. To do so, we want to find a formula for this expression, similar to the one Gauss found for the expression $\sum_{k=1}^n k$ (you can see problem 1).

Hint: thanks to a computer, calculate the sum $\sum_{k=1}^n \frac{1}{k(k+1)}$ for values of n ranging from 1 to 10. Look at the results and try to conjecture a formula for calculating this sum. Then, prove by induction that the formula holds for every natural number $n \in \mathbb{N}$.
4. Prove by induction that the number $n^3 + 2n$ is a multiple of 3 for every natural number $n \in \mathbb{N}$.
5. Although the irrational number $\sqrt{2}$ cannot be expressed as a fraction, we can place it on a number line (in fact, we found it between 1,4 and 1,5) using only a ruler and a compass. How exactly would you place it?
6. Find the solutions of the following quadratic equation: $x^2 - 2x + 5 = 0$. Do you observe any common feature in its solutions?
7. Find the solutions of the following quadratic equation: $x^2 - 6x + 25 = 0$. Do you observe any common feature in the solutions of the equations you have solved?
8. Represent the following numbers in the complex plane: $z_1 = 3$, $z_2 = 3i$, $z_3 = -2i$, $z_4 = 1 + 2i$, $z_5 = -2 + i$.
9. Add the complex numbers $2 - i$ and $1 + 3i$ graphically.
10. Consider the complex numbers $z_1 = 3 + 4i$ and $z_2 = 1 - 2i$. Find $z_3 = z_1 + z_2$. Represent these three complex numbers in the complex plane and check that the parallelogram law is satisfied.
11. Consider the complex numbers $z_1 = 3 + 4i$ and $z_2 = 1 - 2i$. Find $z_3 = z_1 - z_2$. Represent these three complex numbers in the complex plane.
12. Consider the complex numbers $z_1 = 3 + 4i$ and $z_2 = 1 - 2i$. Find $z_1 \cdot z_2$.
13. Consider the complex number $z = 4 + 3i$. Find $\bar{z}$. Find now $\bar{\bar{z}}$, what do you notice?
14. Consider $z = 1 + 5i$. Find $\bar{z}$. Find now $z \cdot \bar{z}$; what do you notice?
15. Find the conjugates of the following complex numbers and prove that $z \cdot \bar{z}$ is real and positive (or zero) in all cases: $2 - 5i$, $-4 + 2i$, -5 , $6i$, $x + yi$. Represent the numbers and their conjugates in the complex plane.
16. Consider the complex numbers $z_1 = 3 + 4i$ and $z_2 = 1 - 2i$. Find z_1/z_2 .
17. Find the following complex numbers:

$$\frac{1}{i}, \frac{1}{a+bi}, \frac{1}{2-3i}, \frac{1}{1+i}, \frac{1+2i}{3-4i} + \frac{2-i}{5i}.$$

18. Consider the complex number $-1 - \sqrt{3}i$. Find its distance from the origin and the angle it makes with the positive x -axis. What do you notice?
19. Express the following complex numbers in polar form: $3 + 2i$, $-2 - 5i$, $-4 + 2i$, $4 - 2i$.
20. Express the following complex numbers in polar form: $-i$, $-1 - i$, $1 - \sqrt{3}i$, $-\sqrt{3} + i$, $\sqrt{3} + i$.
21. Express the following complex numbers in standard form: 1_0 , 2_π , $1_{\pi/3}$, $4_{-\pi/6}$, $2_{3\pi/4}$.
22. Express the following complex numbers in polar form: $-2 - 5i$, $-2 + 5i$.
23. Consider the complex numbers $z_1 = 3_{\pi/6}$ and $z_2 = 2_{\pi/4}$. Find: $z_1 + z_2$, $z_1 - z_2$, $z_1 \cdot z_2$, z_1/z_2 .

24. Consider $z = \pi i$, and find e^z expressing the result in polar form. What magnitude and what argument do you find?
25. Consider $z = 2 + \pi i$, and find e^z (bearing in mind that $e^{2+\pi i} = e^2 e^{\pi i}$) expressing the result in polar form. What magnitude and what argument do you find?
26. Express the following complex numbers in exponential form: $\frac{(3-2i)(2+3i)}{3-4i}$, i^{29} , $(1+i)^4$.
27. Consider $z_1 = 1_0$, $z_2 = 1_{\frac{2\pi}{3}}$ and $z_3 = 1_{\frac{4\pi}{3}}$. Find z_1^3 , z_2^3 , and z_3^3 . What do you get?
28. Represent the cube roots of the unity in the complex plane. Connect by straight lines the three roots. What geometric figure do you obtain?
29. Find the square, fourth, fifth and sixth roots of the unity. Represent them in the complex plane. What geometric figures do you obtain?
30. Find the solutions of the following equations: $z^4 + 16 = 0$, $z^6 = i$ and $z^3 + 8i = 0$.

Solutions

1. We are interested in proving the property by induction: $P(n) : 1^3 + 2^3 + \dots + n^3 = \frac{n^2(n+1)^2}{4}$.

a) **Base case:** $P(1)$ holds, i.e., $1^3 = 1$.

b) **Induction step:** Suppose that this induction hypothesis now holds, $1^3 + 2^3 + \dots + n^3 = \frac{n^2(n+1)^2}{4}$ and from here we try to prove that $1^3 + 2^3 + \dots + n^3 + (n+1)^3 = \frac{(n+1)^2((n+1)+1)^2}{4}$. Note that $\frac{(n+1)^2((n+1)+1)^2}{4} = \frac{(n+1)^2(n+2)^2}{4}$. We start, therefore, from $1^3 + 2^3 + \dots + n^3 + (n+1)^3$ and we applied the induction hypothesis in the n first summands, $1^3 + 2^3 + \dots + n^3 + (n+1)^3 = \frac{n^2(n+1)^2}{4} + (n+1)^3$. Calculating in the second member of the equality, we have:

$$\frac{n^2(n+1)^2 + 4(n+1)^3}{4} = \frac{(n+1)^2(n^2 + 4n + 4)}{4} = \frac{(n+1)^2(n+2)^2}{4}$$

Therefore, $1^3 + 2^3 + \dots + n^3 + (n+1)^3 = \frac{(n+1)^2(n+2)^2}{4}$ and the proof is over.

2. We are interested in proving that for all $n \in \mathbb{N}$ the following property holds: $P(n)$: the number $n(n+1)$ is even.

a) **Base case:** if we consider $n = 1$, the number $n(n+1) = 2$ is even, as expected.

b) **Induction step:** Suppose that $n(n+1)$ is even (induction hypothesis) and we want to prove that $(n+1)(n+2)$ is also an even number. The number $(n+1)(n+2)$ can be written as $(n+1)(n+2) = n(n+1) + 2(n+1)$ and we conclude that it is even because it is the sum of two even numbers. The first one is even by the induction hypothesis, and the second one because it has 2 as a factor.

3. First, we will use a computer to calculate the sum $\sum_{k=1}^n \frac{1}{k(k+1)}$ for values of n from 1 to 10. You will see that, for these values, the sum results are:

$$\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{6}, \frac{6}{7}, \frac{7}{8}, \frac{8}{9}, \frac{9}{10}, \frac{10}{11}$$

This leads to the conjecture that $\sum_{k=1}^n \frac{1}{k(k+1)} = \frac{n}{n+1}$. We will prove by induction that this conjecture holds for all $n \in \mathbb{N}$.

a) **Base case:** $P(1)$: consider $n = 1$, $\sum_{k=1}^1 \frac{1}{k(k+1)} = \frac{1}{1 \cdot 2} = \frac{1}{2}$ and $\frac{n}{n+1} = \frac{1}{2}$. Therefore, the formula holds for $n = 1$.

b) **Induction step:** suppose that this induction hypothesis now holds: $P(n) : \sum_{k=1}^n \frac{1}{k(k+1)} = \frac{n}{n+1}$. From here we try to prove that $\sum_{k=1}^{n+1} \frac{1}{k(k+1)} = \frac{n+1}{(n+1)+1}$. To achieve this, we use the induction hypothesis and replace the n first summands of the first member by $n/(n+1)$:

$$\begin{aligned} \sum_{k=1}^{n+1} \frac{1}{k(k+1)} &= \sum_{k=1}^n \frac{1}{k(k+1)} + \frac{1}{(n+1)((n+1)+1)} = \\ &= \frac{n}{n+1} + \frac{1}{(n+1)((n+1)+1)} = \frac{n}{n+1} + \frac{1}{(n+1)(n+2)} \end{aligned}$$

If we collect this expression using the common denominator, we obtain:

$$\frac{n}{n+1} + \frac{1}{(n+1)(n+2)} = \frac{n(n+2)}{(n+1)(n+2)} + \frac{1}{(n+1)(n+2)} = \frac{n(n+2)+1}{(n+1)(n+2)} = \frac{n^2+2n+1}{(n+1)(n+2)}$$

Finally, if we observe that $n^2 + 2n + 1 = (n+1)^2$ and we simplify, we have:

$$\frac{n^2+2n+1}{(n+1)(n+2)} = \frac{(n+1)^2}{(n+1)(n+2)} = \frac{(n+1)}{(n+2)}$$

Summarizing, we have seen that:

$$\sum_{k=1}^{n+1} \frac{1}{k(k+1)} = \frac{(n+1)}{(n+2)} = \frac{(n+1)}{(n+1)+1}$$

Therefore, the formula also holds for $n+1$, and so ends the proof by induction.

4. We will prove by induction that for all $n \in \mathbb{N}$ the following property holds: $P(n) : n^3 + 2n$ is a multiple of 3.

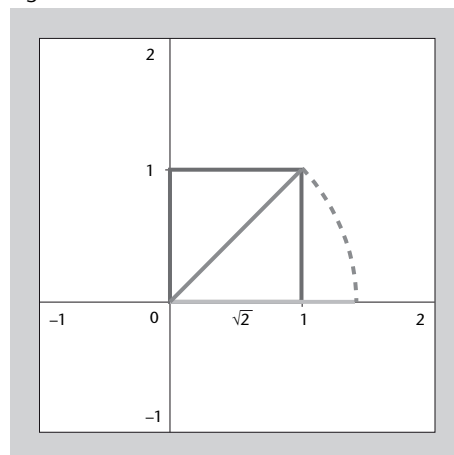
a) **Base case:** $P(1) : 1^3 + 2 \cdot 1 = 3$ is a multiple of 3.

b) **Induction step.** Induction hypothesis: suppose that $n^3 + 2n$ is a multiple of 3. It must be proven: $P(n+1) : (n+1)^3 + 2(n+1)$ is a multiple of 3.

Note that $(n+1)^3 + 2(n+1) = n^3 + 3n^2 + 3n + 1 + 2n + 2 = (n^3 + 2n) + 3(n^2 + n + 1)$ and this number is a multiple of 3, since it is the sum of two multiples of 3: the first is by the induction hypothesis and the second because the factor 3 appears multiplying the expression $n^2 + n + 1$, which is an integer.

5. Let us consider a square of side 1. Applying the Pythagorean theorem, we know that its diagonal has length $d = \sqrt{1^2 + 1^2} = \sqrt{2}$. Therefore, to place the number $\sqrt{2}$ on the real line it is only necessary to have a ruler, a square and a compass. Thanks to the ruler and the square, we draw two coordinate axes and on these axes we draw the square $[0,1] \times [0,1]$. Once the square is drawn, we draw its diagonal. Finally, with the tip of the compass at the point $(0,0)$ we move the diagonal of the square on the x -axis. See figure 18.

Figure 18



6. In this exercise, we must calculate the solutions of the equation $x^2 - 2x + 5 = 0$, and we know that if when calculating them we get a root of a negative number, we can assume that $\sqrt{-1} = i$. To calculate the solutions, we will use the formula that allows us to calculate the solutions of an equation of the form $ax^2 + bx + c = 0$. In our case $a = 1$, $b = -2$ and $c = 5$; therefore:

$$\begin{aligned} x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{2 \pm \sqrt{4 - 20}}{2} \\ &= \frac{2 \pm \sqrt{-16}}{2} = 1 \pm \frac{\sqrt{16}\sqrt{-1}}{2} = 1 \pm \frac{4i}{2} = 1 \pm 2i \end{aligned}$$

Therefore, the two solutions are $1 + 2i$ and $1 - 2i$. It can be seen that the two solutions are the same except for the constant that accompanies the i , which changes its sign.

7. This exercise is the same as the previous exercise, but the coefficients of the equation we want to solve change. As in the previous exercise, we will use the formula that allows us to calculate the solutions of an equation of the form $ax^2 + bx + c = 0$. Now $a = 1$, $b = -6$ and $c = 25$; therefore:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{6 \pm \sqrt{36 - 100}}{2}$$

$$= \frac{6 \pm \sqrt{-64}}{2} = 3 \pm \frac{\sqrt{64}\sqrt{-1}}{2} = 3 \pm \frac{8i}{2} = 3 \pm 4i.$$

Therefore, the two solutions are $3 + 4i$ and $3 - 4i$. Notice that the two solutions are equal except for the constant that accompanies i , which changes its sign. That is, the solutions are of the form:

$$x = c_1 + c_2 i \quad y \quad x = c_1 - c_2 i$$

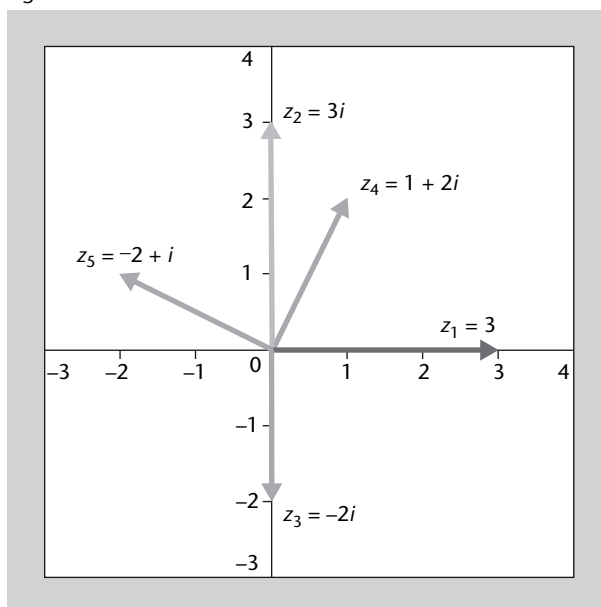
in which the constants c_1 and c_2 depend on the equation under consideration.

8. Recall that a complex number $a + bi$ can be associated with a point in the complex plane, the point (a, b) . In this way, the points we have been given can be associated with the points:

$$\begin{aligned} z_1 = 3 & \rightarrow (3, 0) \\ z_2 = 3i & \rightarrow (0, 3) \\ z_3 = -2i & \rightarrow (0, -2) \\ z_4 = 1 + 2i & \rightarrow (1, 2) \\ z_5 = -2 + i & \rightarrow (-2, 1) \end{aligned}$$

We have represented these points in the complex plane in figure 19.

Figure 19

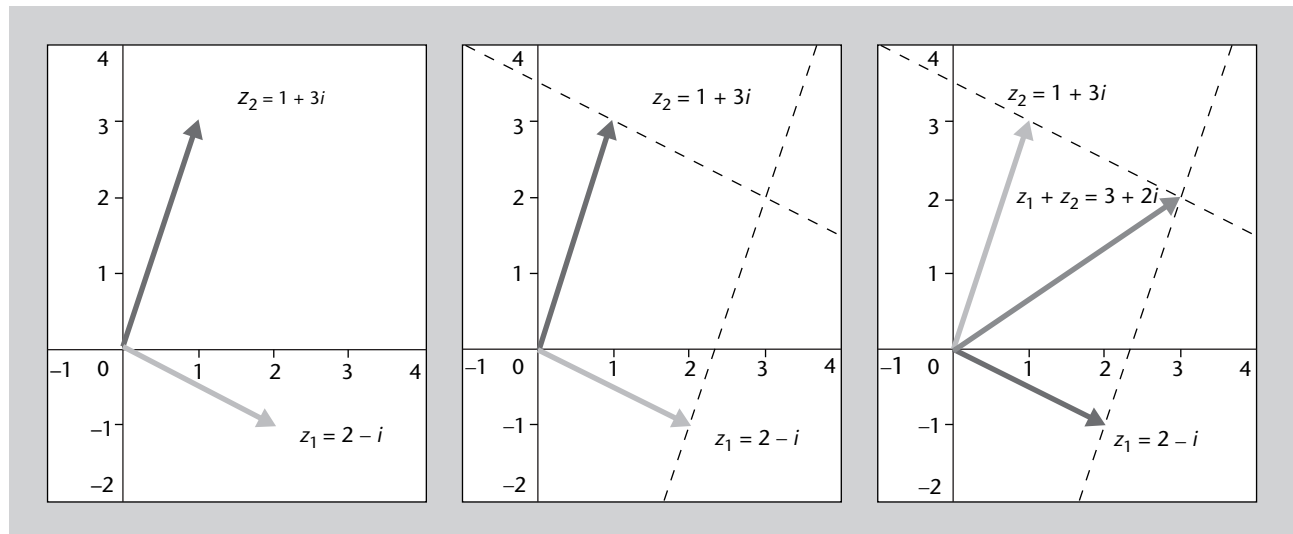


9. To draw the sum of the two numbers $2 - i$ and $1 + 3i$, the first thing we must do is to represent them graphically (you can see the plot on the left of figure 20). These points are associated to the vectors $(2, -1)$ and $(1, 3)$.

The second step is to draw two lines parallel to the vectors so that both vectors and both parallels form a parallelogram (as seen in the middle figure of the plot).

Finally, the sum of both vectors has as origin the point $(0, 0)$ and as end the intersection of the two drawn straight lines (you can see figure 20, on the right).

Figure 20



10. Let us consider the complex numbers $z_1 = 3 + 4i$ and $z_2 = 1 - 2i$. We must calculate their sum $z_1 + z_2$. Recall that to add two complex numbers, we must add the two real parts and the two imaginary parts separately (it is as if we consider the i as a variable and operate with the numbers).

$$z_1 + z_2 = (3 + 4i) + (1 - 2i) = (3 + 1) + (4 - 2)i$$

$$= 4 + 2i$$

To represent these numbers in the complex plane, we must remember that every complex number of the form $a + bi$ is associated with the point (a, b) on the complex plane. Therefore, we must represent the points:

Complex number	→	Point on the plane
$z_1 = 3 + 4i$	→	$(3, 4)$
$z_2 = 1 - 2i$	→	$(1, -2)$
$z_1 + z_2 = 4 + 2i$	→	$(4, 2)$

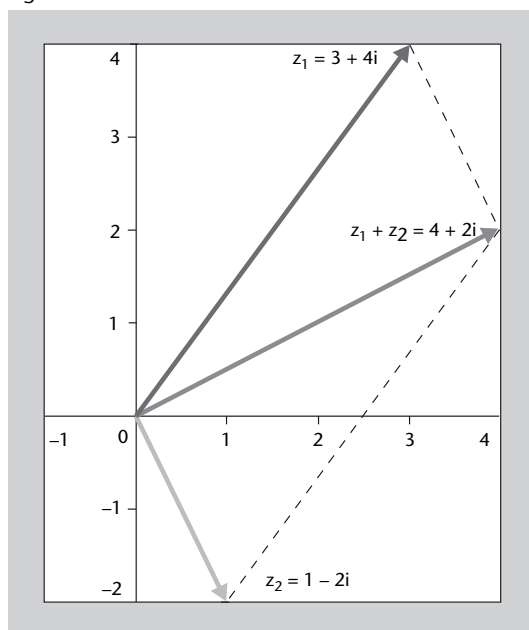
See figure 21.

11. Let us consider the complex numbers $z_1 = 3 + 4i$ and $z_2 = 1 - 2i$. We must calculate their subtraction $z_1 - z_2$. Recall that to subtract two complex numbers, we must subtract the two real parts and the two imaginary parts separately (it is as if we consider the i as a variable and operate with the numbers).

$$z_1 - z_2 = (3 + 4i) - (1 - 2i) = (3 - 1) + (4 - (-2))i$$

$$= 2 + 6i$$

Figure 21

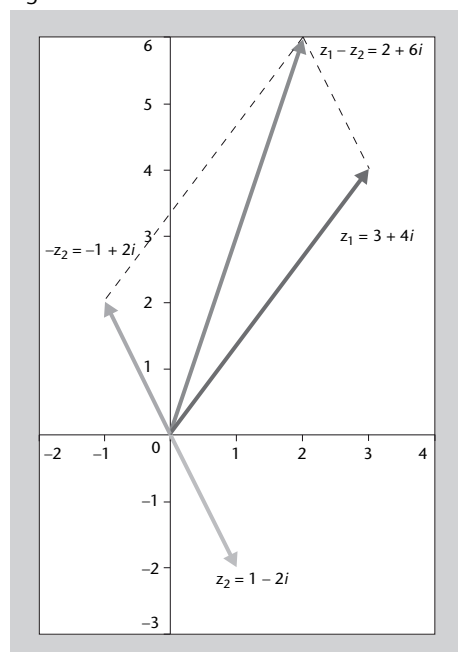


To represent these numbers in the complex plane, we must remember that every complex number of the form $a + bi$ is associated with the point (a, b) on the complex plane. Therefore, we must represent the points:

Complex number	→	Point on the plane
$z_1 = 3 + 4i$	→	(3, 4)
$z_2 = 1 - 2i$	→	(1, -2)
$z_1 - z_2 = 2 + 6i$	→	(2, 6)

Note that to subtract two complex numbers z_1 and z_2 what we do is, basically, to add the numbers z_1 and $-z_2$, as shown in figure 22.

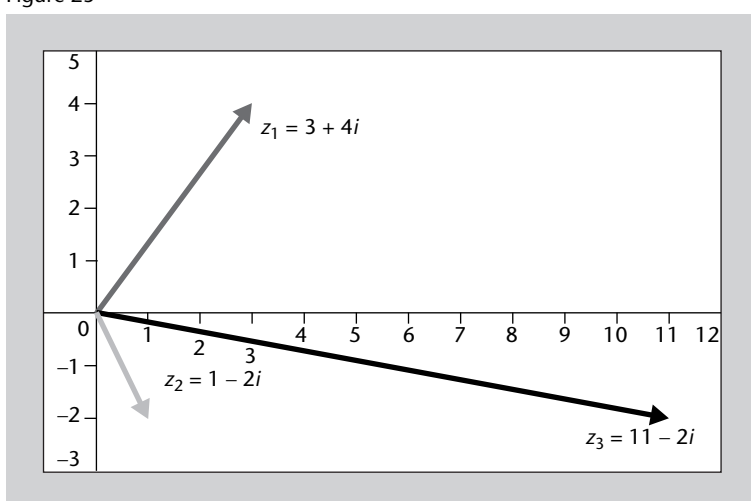
Figure 22



12. Remember that the product of two complex numbers is performed in the same way that we would perform the product of two polynomials of the form $(a + bx) \cdot (c + dx) = a(c + dx) + bx(c + dx) = ac + adx + bcx + bdx^2$. The only thing we must consider is that when we get a i^2 we must substitute it by -1 (figure 23).

$$\begin{aligned}
 z_1 \cdot z_2 &= (3 + 4i) \cdot (1 - 2i) \\
 &= 3 \cdot (1 - 2i) + 4i \cdot (1 - 2i) \\
 &= 3 \cdot 1 + 3 \cdot (-2)i + 4i \cdot 1 + 4i \cdot (-2i) \\
 &= 3 - 6i + 4i - 8i^2 \\
 &= (3 + 8) + (-6 + 4)i \quad (\text{using } i^2 = -1) \\
 &= 11 - 2i
 \end{aligned}$$

Figure 23



13. Consider $z = 4 + 3i$. The conjugate of this number is obtained by changing the sign of the imaginary part, that is:

$$z = 4 + 3i \longrightarrow \bar{z} = 4 - 3i$$

Let us observe that if we calculate the conjugate again, i.e., if we calculate $\bar{\bar{z}}$ we have:

$$z = 4 + 3i \longrightarrow \bar{z} = 4 - 3i \longrightarrow \bar{\bar{z}} = 4 - (-3)i = 4 + 3i$$

That is to say, $\bar{\bar{z}} = z$.

14. Consider $z = 1 + 5i$, and therefore $\bar{z} = 1 - 5i$. Now we calculate $z \cdot \bar{z}$.

$$\begin{aligned}
 z \cdot \bar{z} &= (1 + 5i) \cdot (1 - 5i) \\
 &= 1 \cdot 1 + 1 \cdot (-5)i + 5i \cdot 1 + 5i \cdot (-5i) \\
 &= 1 - 5i + 5i - 25i^2 \\
 &= 1 - 25(-1) \\
 &= 1 + 25 = 26
 \end{aligned}$$

We see, then, that $z \cdot \bar{z}$ is a positive real number, and furthermore we observe:

$$z \cdot \bar{z} = \operatorname{Re}(z)^2 + \operatorname{Im}(z)^2 = (1)^2 + (5^2) = 1 + 25 = 26.$$

15. Let us first consider $z_1 = 2 - 5i$. The conjugate of a complex number is obtained by changing the sign of the imaginary part, i.e.:

$$z_1 = 2 - 5i \longrightarrow \bar{z}_1 = 2 + 5i$$

Then:

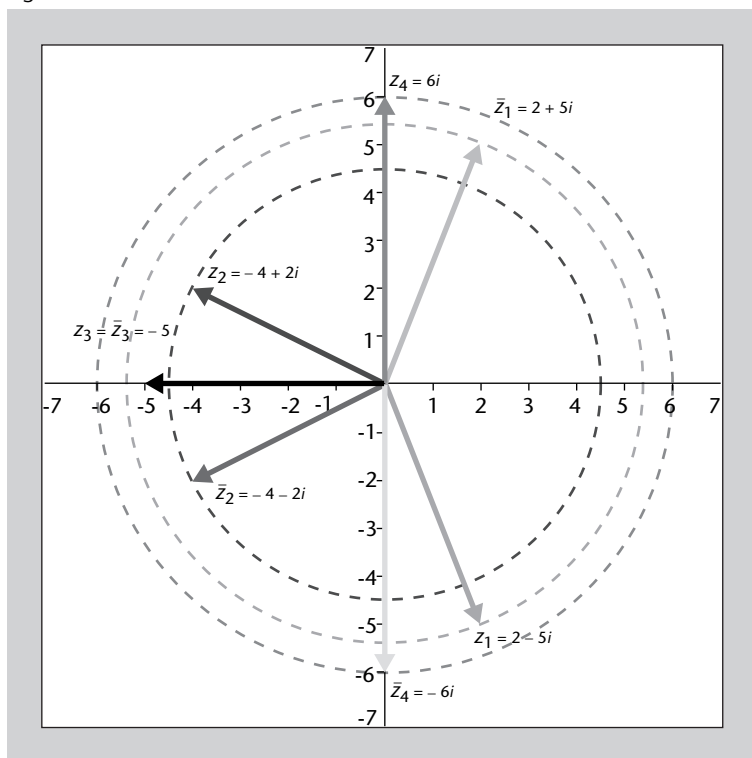
$$\begin{aligned} z_1 \cdot \bar{z}_1 &= (2 - 5i) \cdot (2 + 5i) = 2 \cdot 2 + 2 \cdot 5i - 5i \cdot 2 - 5i \cdot 5i \\ &= 4 + 10i - 10i - 25i^2 \\ &= 4 - 25(-1) = 4 + 25 = 29 \end{aligned}$$

Notice that when we make the product of z_1 by its conjugate, we have:

$$z_1 \cdot \bar{z}_1 = 4 + 25 = \operatorname{Re}(z_1)^2 + \operatorname{Im}(z_1)^2 = 2^2 + (-5)^2 = 4 + 25 = 29$$

The graphical representation of z_1 and $\bar{z}_1$ can be seen in figure 24.

Figure 24



We will do the same with $z_2 = -4 + 2i$. The conjugate of a complex number is obtained by changing the sign of the imaginary part, that is:

$$z_2 = -4 + 2i \longrightarrow \bar{z}_2 = -4 - 2i$$

Then:

$$\begin{aligned}
 z_2 \cdot \bar{z}_2 &= (-4 + 2i) \cdot (-4 - 2i) = (-4) \cdot (-4) + (-4) \cdot (-2i) + 2i \cdot (-4) + 2i \cdot (-2i) \\
 &= 16 + 8i - 8i - 4i^2 \\
 &= 16 - 4(-1) = 16 + 4 = 20
 \end{aligned}$$

Notice that when we calculate the product of z_2 by its conjugate, we have the same result as in the previous example:

$$z_2 \cdot \bar{z}_2 = 16 + 4 = \operatorname{Re}(z_2)^2 + \operatorname{Im}(z_2)^2 = (-4)^2 + (-2)^2 = 16 + 4 = 20$$

The graphical representation of z_2 and $\bar{z}_2$ can be seen in figure 24.

Let us now consider $z_3 = -5$, i.e., a real number, since the imaginary part of z_3 is zero. In this case:

$$z_3 = -5 \longrightarrow \bar{z}_3 = z_3 = -5$$

Then:

$$z_3 \cdot \bar{z}_3 = (-5) \cdot (-5) = 25$$

Note that also in this case we could have used the formula:

$$z_3 \cdot \bar{z}_3 = \operatorname{Re}(z_3)^2 + \operatorname{Im}(z_3)^2 = (-5)^2 + 0^2 = 25$$

The representation of z_3 and $\bar{z}_3$ can be seen, in the same way, in figure 24.

Let us now consider $z_4 = 6i$, i.e., a pure imaginary number, since the real part of z_4 is zero. In this case:

$$z_4 = 6i \longrightarrow \bar{z}_4 = -z_4 = -6i$$

Then:

$$z_4 \cdot \bar{z}_4 = (6i) \cdot (-6i) = -36i^2 = 36$$

Note that also in this case we could have used the formula:

$$z_4 \cdot \bar{z}_4 = \operatorname{Re}(z_4)^2 + \operatorname{Im}(z_4)^2 = 0^2 + 6^2 = 36$$

The graphical representation of z_3 and $\bar{z}_3$ can be seen in figure 24.

Finally, we will do the exercise for a generic complex number $z_5 = x + yi$. In this case:

$$z_5 = x + yi \longrightarrow \bar{z}_5 = x - yi$$

Then:

$$\begin{aligned} z_5 \cdot \bar{z}_5 &= (x + yi) \cdot (x - yi) = x \cdot x + x \cdot (-yi) + yi \cdot x + yi \cdot (-yi) \\ &= x^2 - xyi + xyi - y^2 i^2 \\ &= x^2 - y^2 i^2 = x^2 + y^2 \end{aligned}$$

Note, as we have already seen above, that indeed:

$$z_5 \cdot \bar{z}_5 = x^2 + y^2 = \operatorname{Re}(z_5)^2 + \operatorname{Im}(z_5)^2$$

16. Remember that when we have a quotient of complex numbers, the first thing we must do is to get i out of the denominator. To achieve this, we must multiply by the conjugate of the complex number in the denominator.

$$\begin{aligned} \frac{z_1}{z_2} &= \frac{3 + 4i}{1 - 2i} = \frac{3 + 4i}{1 - 2i} \cdot \frac{1 + 2i}{1 + 2i} = \frac{(3 + 4i)(1 + 2i)}{(1 - 2i)(1 + 2i)} \\ &= \frac{3 + 6i + 4i + 8i^2}{1 + 2i - 2i - 4i^2} = \frac{3 - 8 + 10i}{1 + 4} = \frac{-5 + 10i}{5} = -1 + 2i \end{aligned}$$

17. In this exercise, we must calculate quotients of complex numbers. As in the previous exercise, remember that when we have a quotient of complex numbers, the first thing we must do is to get that there is no i left in the denominator. To achieve this, we must multiply by the conjugate of the complex number in the denominator. We start with the first number $1/i$:

$$\frac{1}{i} = \frac{1}{i} \cdot \frac{-i}{-i} = \frac{-i}{-i^2} = \frac{-i}{1} = -i$$

The quotient $1/(a + bi)$ is calculated similarly:

$$\begin{aligned} \frac{1}{a + bi} &= \frac{1}{a + bi} \cdot \frac{a - bi}{a - bi} = \frac{a - bi}{(a + bi)(a - bi)} \\ &= \frac{a - bi}{a^2 - abi + abi - b^2 i^2} = \frac{a - bi}{a^2 - b^2(-1)} = \frac{a - bi}{a^2 + b^2} \end{aligned}$$

To perform the following calculation, there are two different options, since we have a product of complex numbers in the denominator. The first option is to do the first product of the two numbers from below and then multiply by the conjugate of the result. The second option is to multiply by the conjugates of the two numbers in the denominators and then operate. The first option would result in:

$$\begin{aligned} \frac{1}{2 - 3i} \cdot \frac{1}{1 + i} &= \frac{1}{(2 - 3i)(1 + i)} = \frac{1}{2 + 2i - 3i - 3i^2} = \frac{1}{2 - i + 3} = \frac{1}{5 - i} \\ &= \frac{1}{5 - i} \cdot \frac{5 + i}{5 + i} = \frac{5 + i}{(5 - i)(5 + i)} = \frac{5 + i}{25 + 5i - 5i - i^2} = \frac{5 + i}{26} \end{aligned}$$

We will see how it would be done with the second option:

$$\begin{aligned}\frac{1}{2-3i} \cdot \frac{1}{1+i} &= \frac{1}{2-3i} \cdot \frac{2+3i}{2+3i} \cdot \frac{1}{1+i} \cdot \frac{1-i}{1-i} = \frac{2+3i}{(2-3i)(2+3i)} \cdot \frac{1-i}{(1+i)(1-i)} \\ &= \frac{2+3i}{4+9} \cdot \frac{1-i}{1+1} = \frac{(2+3i)(1-i)}{26} \\ &= \frac{2-2i+3i-3i^2}{26} = \frac{2+i+3}{26} = \frac{5+i}{26}\end{aligned}$$

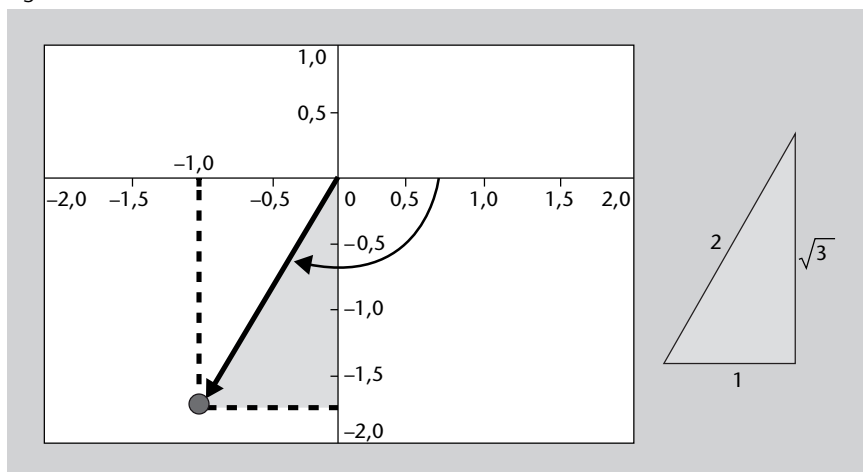
Note that in this exercise, we have assumed that a number by its conjugate is the square of the real part plus the square of the imaginary part. That is, that $(2-3i)(2+3i) = 4+6i-6i-9i^2 = 4+9 = 2^2 + (-3)^2$ and that $(1+i)(1-i) = 1-i+i-i^2 = 1+1 = 1^2 + 1^2$.

Finally, we compute the result of the following operation: $\frac{1+2i}{3-4i} + \frac{2-i}{5i}$. In this case, we must first calculate each of the terms to be added separately and at the end we must add up the results.

$$\begin{aligned}\frac{1+2i}{3-4i} + \frac{2-i}{5i} &= \frac{1+2i}{3-4i} \cdot \frac{3+4i}{3+4i} + \frac{2-i}{5i} \cdot \frac{-5i}{-5i} = \frac{(1+2i)(3+4i)}{(3-4i)(3+4i)} + \frac{(2-i)(-5i)}{-25i^2} \\ &= \frac{(1+2i)(3+4i)}{3^2+4^2} + \frac{(2-i)(-5i)}{25} = \frac{3+4i+6i+8i^2}{25} + \frac{-10i+5i^2}{25} \\ &= \frac{3+10i-8}{25} + \frac{-10i-5}{25} = \frac{-5+10i}{25} + \frac{-10i-5}{25} \\ &= \frac{-1+2i}{5} + \frac{-2i-1}{5} = \frac{-1+2i-2i-1}{5} = \frac{-2}{5} = -\frac{2}{5}\end{aligned}$$

18. Let us consider first the number $z = -1 - \sqrt{3}i$. To calculate its distance from the origin and the angle it makes with the positive x -axis, we will first draw a picture. The number $z = -1 - \sqrt{3}i$ is associated to the point $(-1, -\sqrt{3})$ as seen in figure 25; therefore, it is a number that lies in the third quadrant.

Figure 25



We calculate the distance to the origin using the Pythagorean theorem. Notice that the point $(-1, \sqrt{3})$ determines a right triangle of legs with lengths 1 and $\sqrt{3}$, and what we want to do is to calculate the length of the hypotenuse. Therefore:

$$r = \sqrt{1^2 + (\sqrt{3})^2} = \sqrt{1+3} = \sqrt{4} = 2$$

To calculate the angle it forms with the x -axis, we will first calculate the angle we have marked in the right triangle of figure 25. Since it is a right triangle, we know that

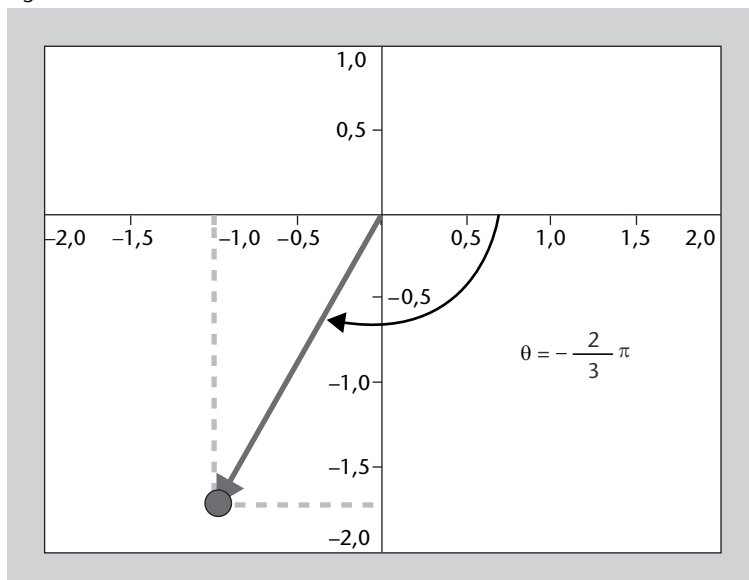
$$\sin(\theta) = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{6}$$

The angle we are looking for is the angle that we have marked in the triangle plus $\pi/2$ rad and, in addition, since it goes clockwise, we must change its sign. Therefore, the angle we are looking for is:

$$-\left(\frac{\pi}{6} + \frac{\pi}{2}\right) = -\frac{\pi + 3\pi}{6} = -\frac{4\pi}{6} = -\frac{2\pi}{3}$$

Finally, we draw a picture of the results we have obtained, which you will find in figure 26.

Figure 26



19. Let us first consider the number $z = 3 + 2i$. When we want to convert a number from standard form to polar form, it is crucial, in order not to make a mistake in the result, to make a plot. Therefore, the first thing we do is to draw the number $z = 3 + 2i$ in the complex plane. This number is associated to the point $(3,2)$, as seen in figure 27; therefore, it is a number that lies in the first quadrant.

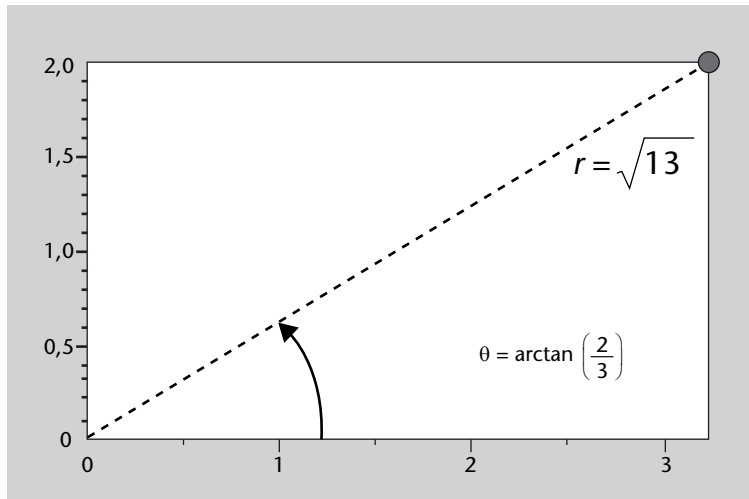
From the plot, we can see that the argument (or angle of this complex number) must be between 0 and $\pi/2$ rad. Now we will calculate the modulus and the argument. The modulus (distance from the point $(3,2)$ to the origin) is calculated as follows:

$$r = \sqrt{3^2 + 2^2} = \sqrt{9 + 4} = \sqrt{13}$$

To calculate the argument, θ , we observe that from the plot we have:

$$\tan(\theta) = \frac{\sin(\theta)}{\cos(\theta)} = \frac{2/r}{3/r} = \frac{2}{3}$$

Figure 27



Therefore, we obtain the angle θ as $\theta = \arctan\left(\frac{2}{3}\right)$. If we calculate this value with the calculator (in rad mode), we have that $\arctan\left(\frac{2}{3}\right) \approx 0,5880$ rad, which is the value we are looking for, since:

$$0 \leq 0,588 \leq \frac{\pi}{2} \approx 1,5708$$

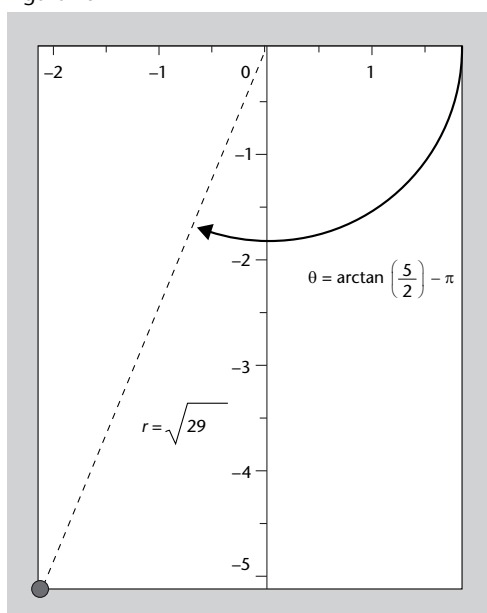
Therefore:

$$z = 3 + 2i = \sqrt{13}_{0,5880 \text{ rad}} = \sqrt{13}_{0,5880 \text{ rad} \cdot \frac{180^\circ}{\pi \text{ rad}}} = 3,61_{33,69^\circ}$$

With the rest of the points, we proceed in the same way. First we make a plot to situate in which quadrant we lie, and then we calculate the modulus and the argument.

Consider now $z = -2 - 5i$. This complex number is associated with the point $(-2, -5)$ as seen in figure 28; therefore, it is a number that lies in the third quadrant.

Figure 28



From the plot, we can see that the argument (or angle) of this complex number must be found to be between $180^\circ = \pi$ rad and $270^\circ = 3\pi/2$ rad. Note that we can also say that the angle lies between $-90^\circ = -\pi/2$ rad and $-180^\circ = -\pi$ rad. The magnitude is:

$$r = \sqrt{(-2)^2 + (-5)^2} = \sqrt{4 + 25} = \sqrt{29}$$

To calculate the argument θ we observe that from the plot we have:

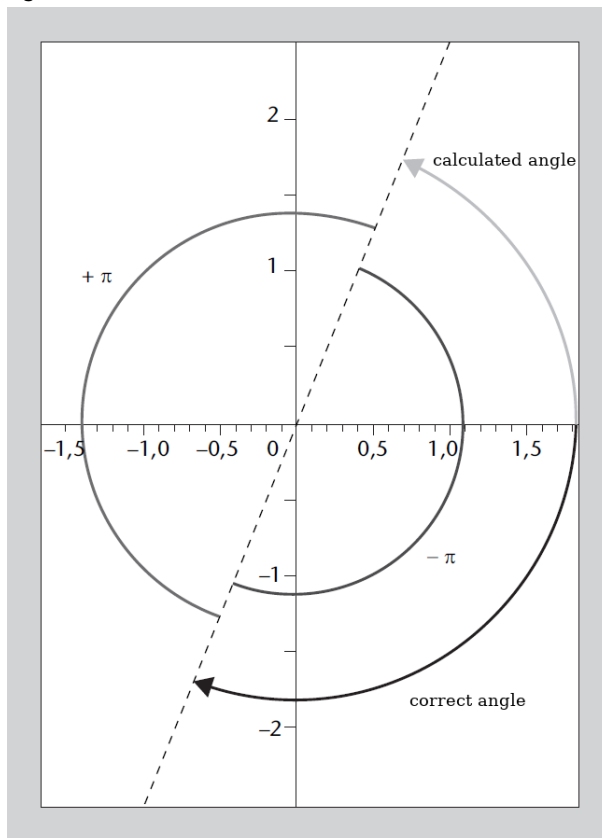
$$\tan(\theta) = \frac{\sin(\theta)}{\cos(\theta)} = \frac{-5/r}{-2/r} = \frac{-5}{-2} = \frac{5}{2}$$

Therefore, we obtain the angle θ as $\theta = \arctan\left(\frac{5}{2}\right)$. If we calculate this value with the calculator (in `rad` mode) we have that $\arctan\left(\frac{5}{2}\right) = 1,1903$ rad, which is not the value we are looking for, since:

$$1,1903 \leq \pi = 3,1416$$

You can see figure 29, in which we have represented the angle that we want to and the angle that the calculator gives us.

Figure 29



The angle we have obtained is the equivalent of the first quadrant, as shown in the plot. To retrieve the angle we want from the angle given by the calculator, we must add or subtract π . Although the result will be different, the value obtained will be the same angle. In this case, we have decided to subtract π :

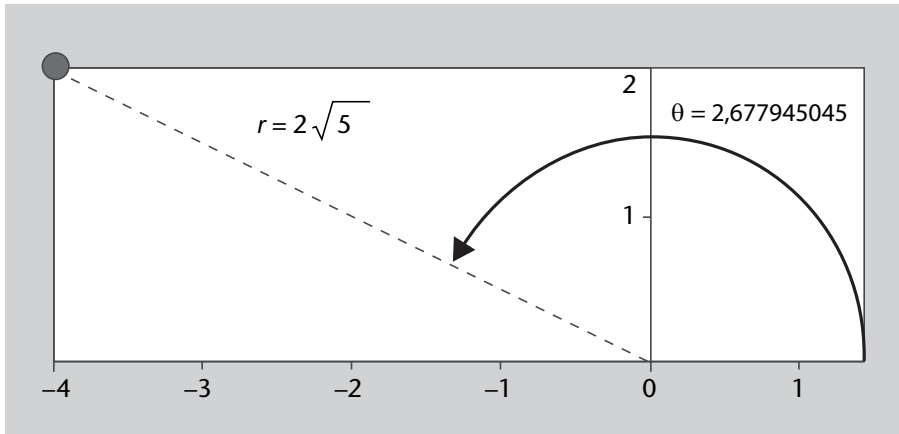
$$\theta = 1,1903 - \pi = -1,9512$$

and the complex number is:

$$z = -2 - 5i = \sqrt{29}_{-1,9512 \text{ rad}} = \sqrt{29}_{-1,9512 \text{ rad} \cdot \frac{180^\circ}{\pi \text{ rad}}} = 5,38_{-111,80^\circ}$$

Let us now consider the number $z = -4 + 2i$. This complex number is associated with the point $(-4,2)$, as can be seen in figure 30; therefore, it is a number that lies in the second quadrant.

Figure 30



From the plot, we can see that the argument (or angle) of this complex number must lie between $90^\circ = \pi/2 \text{ rad}$ and $180^\circ = \pi \text{ rad}$. The magnitude is:

$$r = \sqrt{(-4)^2 + 2^2} = \sqrt{16 + 4} = \sqrt{20} = \sqrt{4} \sqrt{5} = 2\sqrt{5}$$

To calculate the argument θ we observe that from the plot we have:

$$\tan(\theta) = \frac{\sin(\theta)}{\cos(\theta)} = \frac{2/r}{-4/r} = \frac{2}{-4} = -\frac{1}{2}$$

Therefore, we derive the angle θ as

$$\theta = \arctan\left(-\frac{1}{2}\right)$$

If we calculate this value with the calculator (in `rad` mode) we have that:

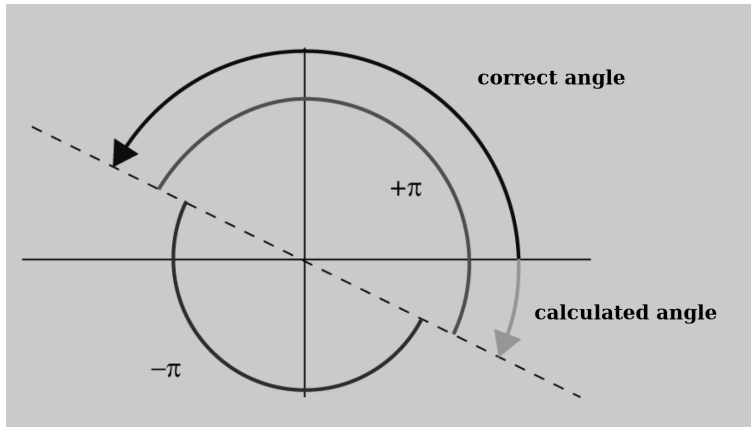
$$\arctan\left(-\frac{1}{2}\right) = -0,4636 \text{ rad}$$

which is not the value we are looking for, since:

$$-0,4636 < 0$$

You can see figure 31, in which we have represented the angle we want to calculate and the angle given by the calculator.

Figure 31



The angle we have obtained is the equivalent of the fourth quadrant, as shown in the plot. To retrieve the angle we want from the angle given by the calculator, we must add or subtract π . In this case, as the calculator gives us a negative value, we add π :

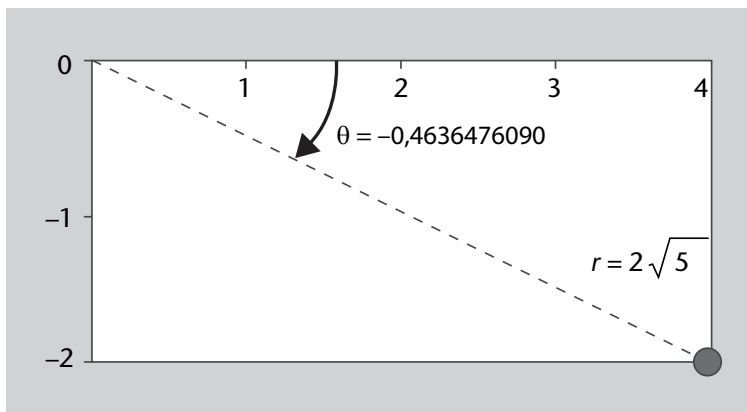
$$\theta = -0,4636 + \pi = 2,6780$$

and the complex number is:

$$z = -4 + 2i = 2\sqrt{5}_{2,6780 \text{ rad}} = 2\sqrt{5}_{2,6780 \text{ rad} \cdot \frac{180^\circ}{\pi \text{ rad}}} = 4,47_{153,44^\circ}$$

Finally, we consider the number $z = 4 - 2i$. This complex number is associated with the point $(4, -2)$, as seen in figure 32; therefore, it is a number that lies in the fourth quadrant.

Figure 32



From the plot, we can see that the argument (or angle) of this complex number must lie between $270^\circ = 3\pi/2 \text{ rad}$ and $360^\circ = 2\pi \text{ rad}$, or between $0^\circ = 0 \text{ rad}$ and $-90^\circ = -\pi/2 \text{ rad}$. The magnitude is:

$$r = \sqrt{4^2 + (-2)^2} = \sqrt{16 + 4} = \sqrt{20} = \sqrt{4 \cdot 5} = 2\sqrt{5}$$

To calculate the argument θ we observe that from the plot we have that:

$$\tan(\theta) = \frac{\sin(\theta)}{\cos(\theta)} = \frac{-2/r}{4/r} = \frac{-2}{4} = -\frac{1}{2}$$

Therefore, the angle θ is obtained as:

$$\theta = \arctan\left(-\frac{1}{2}\right)$$

If we calculate this value with the calculator (in rad mode), we have that:

$$\arctan\left(-\frac{1}{2}\right) = -0,4636 \text{ rad}$$

which is now the angle we are looking for, since:

$$-\frac{\pi}{2} = -1,5708 < -0,4636 < 0$$

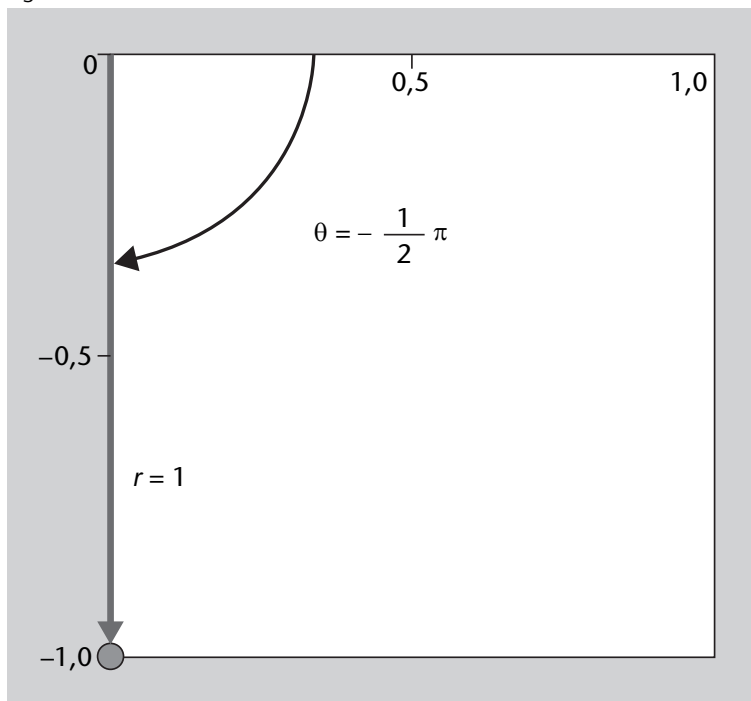
Therefore:

$$z = 4 - 2i = 2\sqrt{5}_{-0,4636 \text{ rad}} = 2\sqrt{5}_{-0,4636 \text{ rad} \cdot \frac{180^\circ}{\pi \text{ rad}}} = 4,47_{-26,56^\circ}$$

20. Remember that when we want to convert a number from standard form to polar form it is vital, in order not to make a mistake in the result, to make a plot. Also, in simple cases, we can give the result directly.

Let us consider first the number $z = -i$, which is associated with the point $(0, -1)$, as shown in figure 33.

Figure 33

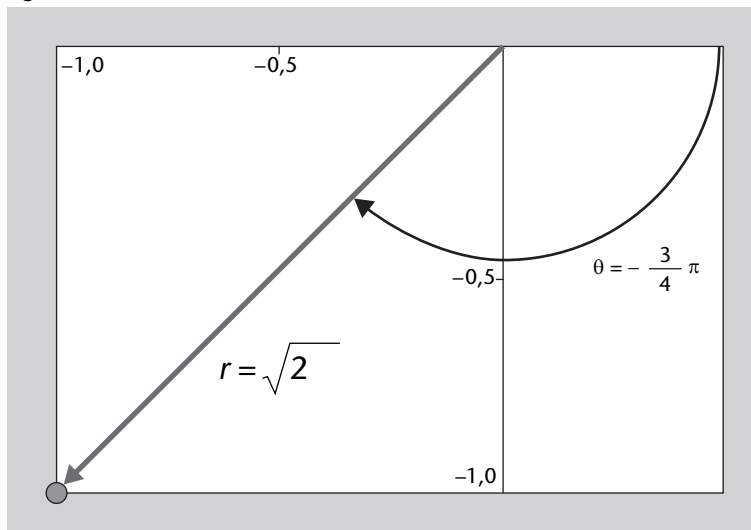


In this case, notice that the distance to the origin is clearly 1 and that the angle is clearly $-90^\circ = -\pi/2$ rad. Therefore, in polar form it is:

$$z = -i = 1_{-\pi/2}$$

Let us consider now the number $z = -1 - i$, which is associated to the point $(-1, -1)$, as shown in figure 34.

Figure 34

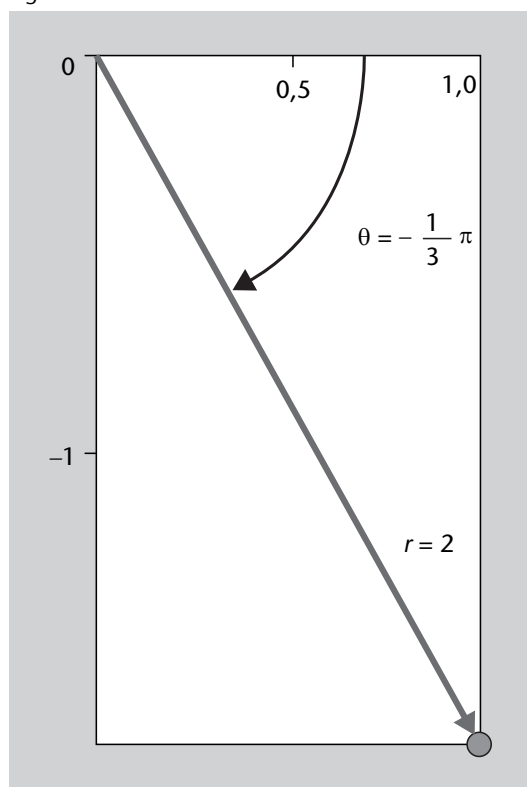


In this case, notice that the distance to the origin is clearly $\sqrt{2}$ because it is the diagonal of a square of side 1. In addition, the angle is clearly $(-90 - 45)^\circ = -135^\circ = -3\pi/4$ rad. Therefore, in polar form it is:

$$z = -1 - i = \sqrt{2}_{-3\pi/4}$$

Let us consider now the number $z = 1 - \sqrt{3}i$, which is associated to the point $(1, -\sqrt{3})$. As seen in figure 35, it is a complex number that lies in the fourth quadrant.

Figure 35



From the plot, we can see that the argument (or angle) of this complex number must lie between $270^\circ = 3\pi/2$ rad and $360^\circ = 2\pi$ rad, or between $0^\circ = 0$ rad and $-90^\circ = -\pi/2$ rad. The magnitude is:

$$r = \sqrt{1^2 + (-\sqrt{3})^2} = \sqrt{1+3} = \sqrt{4} = 2$$

To calculate the argument θ we observe that from the plot we have that:

$$\tan(\theta) = \frac{\sin(\theta)}{\cos(\theta)} = \frac{-\sqrt{3}/r}{1/r} = -\sqrt{3}$$

Therefore, the angle θ is obtained as:

$$\theta = \arctan(-\sqrt{3})$$

If we calculate this value with the calculator (in rad mode) we have that:

$$\arctan(-\sqrt{3}) = -1,047 \text{ rad} = -\frac{\pi}{3} \text{ rad}$$

This angle is the one we are looking for, since:

$$-\frac{\pi}{2} < -\frac{\pi}{3} < 0$$

Therefore:

$$z = 1 - \sqrt{3}i = 2_{-\pi/3 \text{ rad}} = 2_{-\pi/3 \text{ rad} \cdot \frac{180^\circ}{\pi \text{ rad}}} = 2_{-60^\circ}$$

To finish the exercise, we must convert the numbers $z_2 = -\sqrt{3} + i$ and $z_3 = \sqrt{3} + i$ to polar form. We will do this in a simple way with the information we just calculated. Let us consider $z_1 = 1 - \sqrt{3}i$. We have just seen that:

$$z_1 = 1 - \sqrt{3}i = 2_{-\pi/3 \text{ rad}} = 2_{-60^\circ}$$

To finish the exercise, we plot the points z_1 , z_2 and z_3 in the complex plane (see figure 36).

We can see that the three points lie on the same circumference. Therefore, the three points have the same modulus, which is 2.

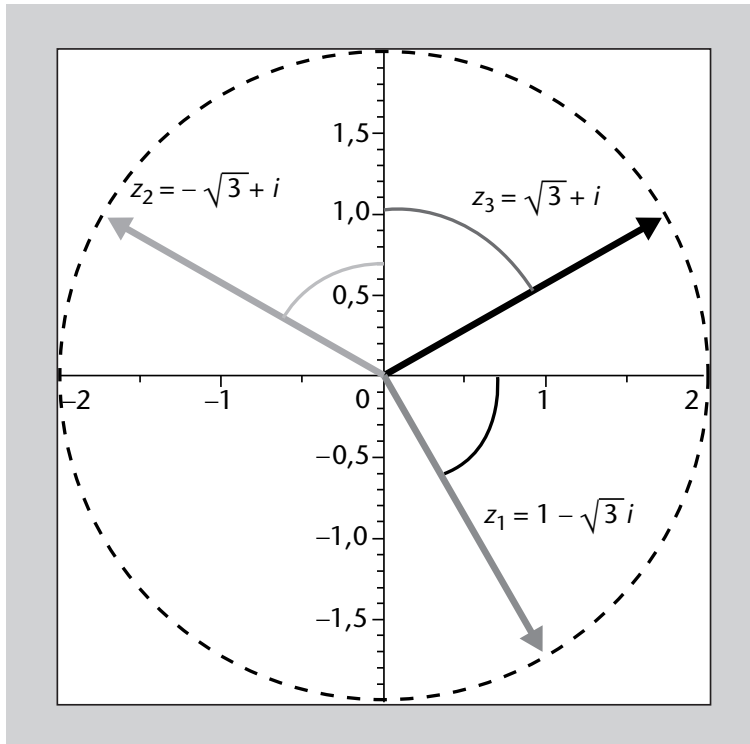
Now we only need to calculate the angles. In figure 36 we have plotted the three angles, which are equal (60°). Therefore, the angle associated with z_2 is $90^\circ + 60^\circ = 150^\circ$. Similarly, the angle associated with z_3 is $90^\circ - 60^\circ = 30^\circ$. Therefore:

$$z_1 = 1 - \sqrt{3}i = 2_{-60^\circ} = 2_{-\pi/3 \text{ rad}}$$

$$z_2 = -\sqrt{3} + i = 2_{150^\circ} = 2_{5\pi/6 \text{ rad}}$$

$$z_3 = \sqrt{3} + i = 2_{30^\circ} = 2_{\pi/6 \text{ rad}}$$

Figure 36



A good exercise to practice is to redo this exercise without the visual aid and check that you obtain the same result.

21. With a view to convert numbers from polar form to standard form, it is not so necessary to make the plot, but it is always advisable to finish checking the results. To do this exercise, we must consider the following formula, which converts from standard to polar form:

$$r_{\theta} = r \cos(\theta) + r \sin(\theta)i$$

Therefore, we have that:

$$z_1 = 1_0 = 1 \cos(0) + 1 \sin(0)i = 1 + 0i = 1$$

$$z_2 = 2_{\pi} = 2 \cos(\pi) + 2 \sin(\pi)i = 2 \cdot (-1) + 0i = -2$$

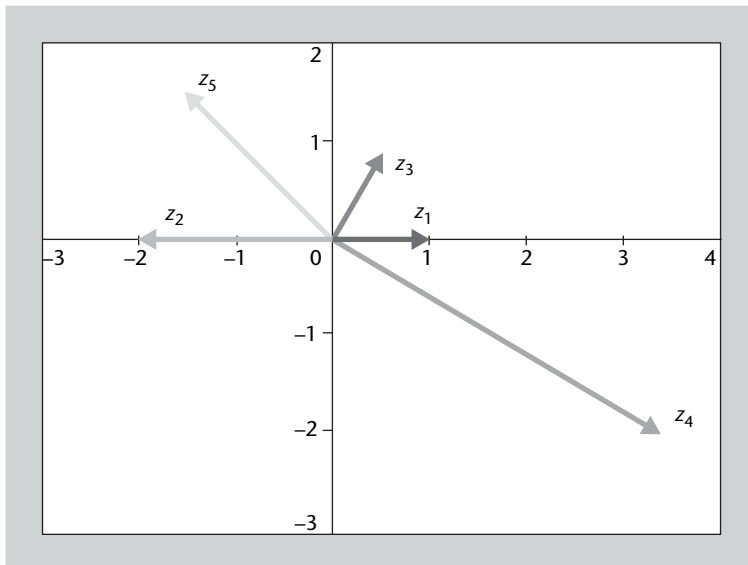
$$z_3 = 1_{\pi/3} = 1 \cos(\pi/3) + 1 \sin(\pi/3)i = 1 \cdot \frac{1}{2} + 1 \cdot \frac{\sqrt{3}}{2}i = \frac{1}{2} + \frac{\sqrt{3}}{2}i$$

$$z_4 = 4_{-\pi/6} = 4 \cos(-\pi/6) + 4 \sin(-\pi/6)i = 4 \cdot \frac{\sqrt{3}}{2} + 4 \cdot \frac{-1}{2}i = 2\sqrt{3} - 2i$$

$$z_5 = 2_{3\pi/4} = 2 \cos(3\pi/4) + 2 \sin(3\pi/4)i = 2 \cdot \frac{-\sqrt{2}}{2} + 2 \cdot \frac{\sqrt{2}}{2}i = -\sqrt{2} + \sqrt{2}i$$

To check the results, we have plotted the numbers on the complex plane (see figure 37).

Figure 37



22. Let us consider the complex numbers $z_1 = -2 - 5i$ and $z_2 = -2 + 5i$. Notice that z_2 is the conjugate of z_1 , i.e.:

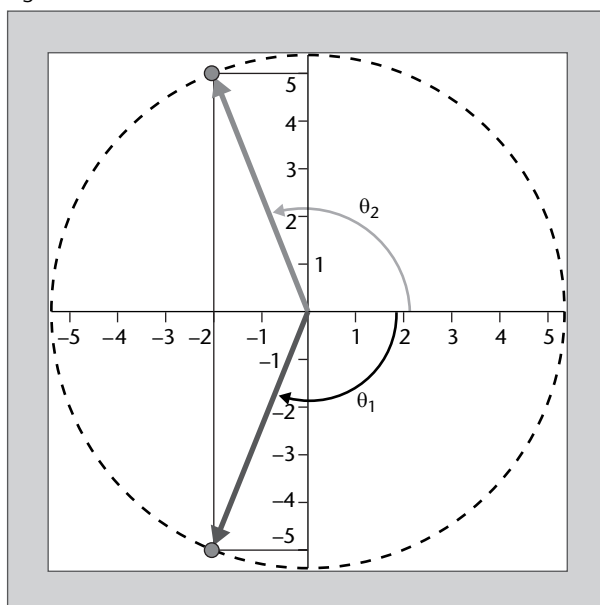
$$z_2 = \bar{z}_1$$

With this information, we already know that the modulus of the two numbers will be the same, which is:

$$r = \sqrt{2^2 + 5^2} = \sqrt{4 + 25} = \sqrt{29}$$

To calculate correctly the angles, what we do first is to make a plot in the complex plane, as you can see in figure 38.

Figure 38



As seen in the plot, the angles of z_1 and z_2 are equal, but the signs are changed. Therefore, if we represent the angle of z_1 as θ_1 and the angle of z_2 as θ_2 we have that:

$$\theta_2 = -\theta_1$$

Note that this is the same as saying that $\theta_2 = 2 - \theta_1$, since we can add (or subtract) 2π to the angles without modifying them.

Therefore, we will now calculate θ_1 . Note that since z_1 lies in the third quadrant, its argument (or angle) must lie between $180^\circ = \pi$ rad and $270^\circ = 3\pi/2$ rad, or between $-90^\circ = -\pi/2$ rad and $-180^\circ = -\pi$ rad. From the plot it can be seen that:

$$\tan(\theta_1) = \frac{\sin(\theta_1)}{\cos(\theta_1)} = \frac{-5/r}{-2/r} = \frac{5}{2}$$

Therefore, the angle θ_1 is obtained as:

$$\theta_1 = \arctan\left(\frac{5}{2}\right)$$

If we calculate this value with the calculator (in rad mode) we have:

$$\arctan\left(\frac{5}{2}\right) = 1,1902 \text{ rad} = 1,1902 \text{ rad} \cdot \frac{180^\circ}{\pi \text{ rad}} = 68,19^\circ$$

Therefore, this is not the angle we are looking for, but we must subtract (or add π), that is, that:

$$\theta_1 = (1,1902 - \pi) \text{ rad} = -1,9514 \text{ rad} = -111,81^\circ$$

Finally:

$$\theta_2 = -\theta_1 = 1,9514 \text{ rad} = 111,81^\circ$$

Therefore:

$$z_1 = -2 - 5i = \sqrt{29}_{-1,9514 \text{ rad}}$$

$$z_2 = -2 + 5i = \sqrt{29}_{1,9514 \text{ rad}}$$

23. Remember that the addition and subtraction of two complex numbers must always be performed in standard form. On the other hand, the product and the division of complex numbers must be performed in standard form as well as in polar or exponential form.

We begin, then, by calculating the sum and subtraction of z_1 and z_2 . Since to add or subtract we must do it in standard form, the first thing we must do is to convert the two numbers to standard form:

$$z_1 = 3_{\pi/6} = 3 \cos(\pi/6) + 3 \sin(\pi/6)i = \frac{3\sqrt{3}}{2} + 3 \frac{1}{2}i = \frac{3\sqrt{3}}{2} + \frac{3}{2}i$$

$$z_2 = 2_{\pi/4} = 2 \cos(\pi/4) + 2 \sin(\pi/4)i = \frac{2\sqrt{2}}{2} + \frac{2\sqrt{2}}{2}i = \sqrt{2} + \sqrt{2}i$$

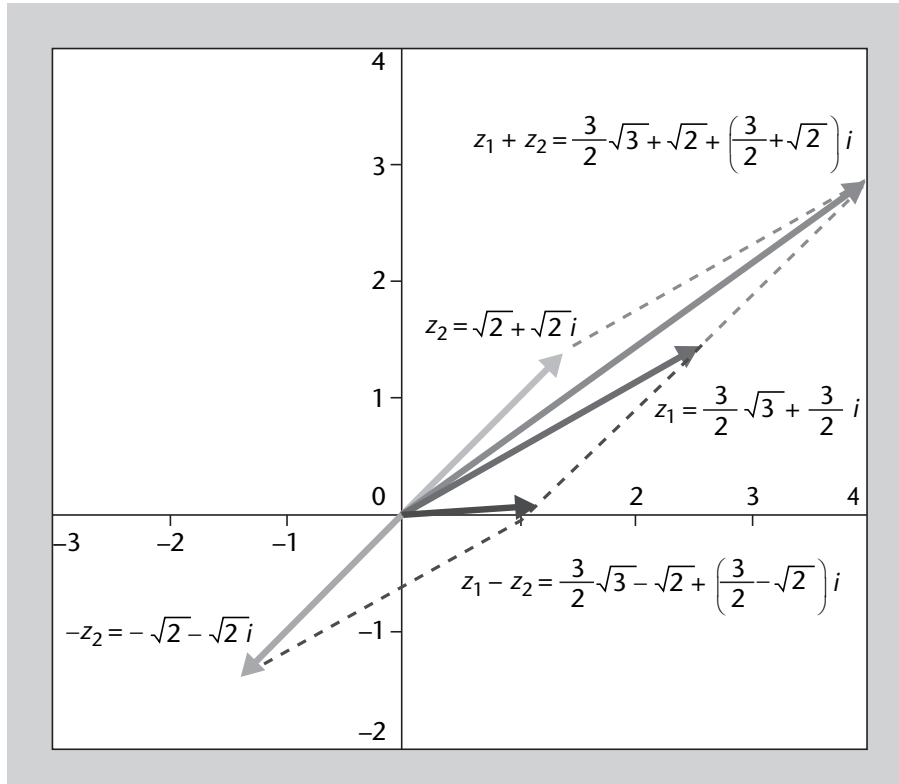
Therefore:

$$z_1 + z_2 = \frac{3\sqrt{3}}{2} + \frac{3}{2}i + \sqrt{2} + \sqrt{2}i = \frac{3\sqrt{3}}{2} + \sqrt{2} + \left(\frac{3}{2} + \sqrt{2}\right)i$$

$$z_1 - z_2 = \frac{3\sqrt{3}}{2} + \frac{3}{2}i - (\sqrt{2} + \sqrt{2}i) = \frac{3\sqrt{3}}{2} - \sqrt{2} + \left(\frac{3}{2} - \sqrt{2}\right)i$$

You can see the results in figure 39.

Figure 39



Now we will calculate the product and the quotient of z_1 and z_2 :

$$z_1 \cdot z_2 = 3_{\pi/6} \cdot 2_{\pi/4} = (3 \cdot 2)_{\pi/6 + \pi/4} = 6_{5\pi/12}$$

$$\frac{z_1}{z_2} = \frac{3_{\pi/6}}{2_{\pi/4}} = \left(\frac{3}{2}\right)_{\pi/6 - \pi/4} = 1,5_{-\pi/12}$$

Note that to calculate the product of two complex numbers, the moduli are multiplied and the arguments are added. Similarly, to calculate the division of two complex numbers, we divide the moduli and subtract the arguments.

24. Using the definition of a pure imaginary complex number, we have:

$$e^{\pi i} = \cos(\pi) + \sin(\pi)i = -1 + 0i = -1$$

Since the result is a negative real number, converting it to polar form is immediate. The modulus of $e^{\pi i}$ is 1 and the angle is π . That is:

$$e^{\pi i} = -1 = 1_{\pi}$$

25. If we consider $z = 2 + \pi i$, we can calculate e^z as:

$$e^z = e^{2+\pi i} = e^2 e^{\pi i}$$

If we now use the result of the previous exercise, we have:

$$e^z = e^2 e^{\pi i} = e^2 \cdot (-1) = -e^2$$

Therefore, again we get a negative real number. Thus:

$$e^z = -e^2 = (e^2)_\pi$$

The modulus of e^z is e^2 and its angle π .

26. To simplify the expressions, we are given in exponential form, the first thing we must do is to convert the numbers we have in standard form to exponential form.

We start by calculating the expression $\frac{(3-2i)(2+3i)}{3-4i}$. To achieve this, we must convert the three complex numbers in the fraction to exponential form.

$$z_1 = 3 - 2i \rightarrow r_1 = \sqrt{3^2 + (-2)^2} = \sqrt{9 + 4} = \sqrt{13}$$

$$\rightarrow \tan(\theta_1) = \frac{-2}{3} \Rightarrow \theta_1 = -0,5880 \text{ rad}$$

$$z_2 = 2 + 3i \rightarrow r_2 = \sqrt{2^2 + 3^2} = \sqrt{4 + 9} = \sqrt{13}$$

$$\rightarrow \tan(\theta_2) = \frac{3}{2} \Rightarrow \theta_2 = 0,9828 \text{ rad}$$

$$z_3 = 3 - 4i \rightarrow r_3 = \sqrt{3^2 + (-4)^2} = \sqrt{9 + 16} = \sqrt{25} = 5$$

$$\rightarrow \tan(\theta_3) = \frac{-4}{3} \Rightarrow \theta_3 = -0,9273 \text{ rad}$$

Therefore, we have that:

$$\begin{aligned} \frac{(3-2i)(2+3i)}{3-4i} &= \frac{z_1 \cdot z_2}{z_3} = \frac{\sqrt{13}e^{-0,5880i} \cdot \sqrt{13}e^{0,9828i}}{5e^{-0,9273i}} \\ &= \frac{\sqrt{13}^2}{5} e^{(-0,5880+0,9828+0,9273)i} = \frac{13}{5} e^{1,3221i} \end{aligned}$$

Finally, we can convert the result to standard form:

$$\begin{aligned} \frac{13}{5} e^{1,3221i} &= \frac{13}{5} \cos(1,3221) + \frac{13}{5} \sin(1,3221)i \\ &= \frac{13}{5} \cdot 0,2461 + \frac{13}{5} \cdot (0,9692)i = 0,6399 + 2,5199i \end{aligned}$$

Note that, since we have approximated the angles to four decimal places, we have obtained an approximation of the result. In fact, it is a good exercise to do the same calculations in standard form and to check the following expression:

$$\frac{(3-2i)(2+3i)}{3-4i} = \frac{16}{25} + \frac{63}{25}i = 0,64 + 2,52i$$

Now we will calculate i^{29} . The number i is associated to the point (0,1) of the complex plane; therefore, it has modulus 1 and angle $\pi/2 = 90$. Therefore, using Moivre's formula we have that:

$$i^{29} = (1e^{\pi/2i})^{29} = (e^{\pi/2i})^{29} = e^{29 \cdot \pi/2i}$$

Note that the obtained angle is:

$$\frac{29}{2}\pi = \frac{29}{4}2\pi = 7,25 \cdot 2\pi$$

which represents more than 7 turns. To know the angle between 0 and 2, we must subtract 7 turns:

$$\frac{29}{2}\pi - 7 \cdot 2\pi = \left(\frac{29}{2} - 14\right)\pi = \frac{29 - 28}{2}\pi = \frac{\pi}{2}$$

Therefore, the result is:

$$e^{\pi/2i} = \cos(\pi/2) + \sin(\pi/2)i = i$$

Finally, we will calculate $(1+i)^4$. First, we will convert the number $1+i$ to polar form. Notice that the complex number $1+i$ is associated to the point (1,1), which has modulus $\sqrt{2}$ and angle $45^\circ = \pi/4$ rad. Therefore:

$$(1+i)^4 = \left(\sqrt{2}e^{(\pi/4)i}\right)^4 = \left(\sqrt{2}\right)^4 e^{4 \cdot (\pi/4)i} = 2^2 e^{\pi i} = 4e^{\pi i}$$

Note that $e^{\pi i}$ is -1 ; therefore:

$$(1+i)^4 = 4e^{\pi i} = -4$$

27. To calculate the cube of z_1 , z_2 and z_3 , we must first convert these numbers to polar form.

$$z_1 = 1_0 = 1$$

$$z_2 = 1_{\frac{2\pi}{3}} = 1e^{2\pi/3i} = e^{2\pi/3i}$$

$$z_3 = 1_{\frac{4\pi}{3}} = 1e^{4\pi/3i} = e^{4\pi/3i}$$

Now we will calculate the cube using the properties of the exponential form. The result will be expressed in standard form:

$$z_1 = 1 \quad \Rightarrow \quad z_1^3 = 1^3 = 1$$

$$z_2 = e^{2\pi/3i} \quad \Rightarrow \quad z_2^3 = (e^{2\pi/3i})^3 = e^{3 \cdot 2\pi/3i} = e^{2\pi i} = \cos(2\pi) + \sin(2\pi)i = 1$$

$$z_3 = e^{4\pi/3i} \quad \Rightarrow \quad z_3^3 = (e^{4\pi/3i})^3 = e^{3 \cdot 4\pi/3i} = e^{4\pi i} = \cos(4\pi) + \sin(4\pi)i = 1$$

Therefore, we see that $z_1^3 = z_2^3 = z_3^3 = 1$.

Remark

Remember that it is very useful for you to always represent the complex numbers you are working with in order not to make mistakes, especially when calculating angles.

28. The three cube roots of unity are:

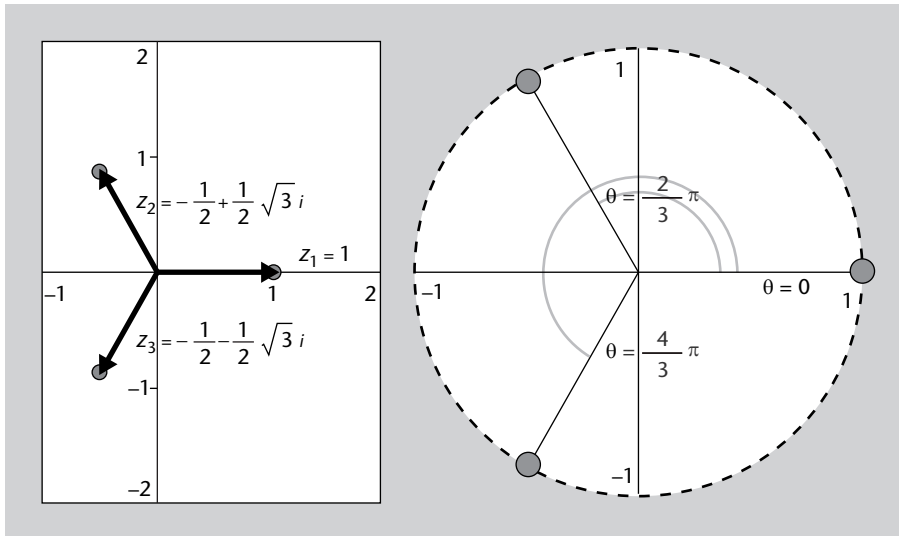
$$z_1 = e^{0i} = 1$$

$$z_2 = e^{2\pi/3i} = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$$

$$z_3 = e^{4\pi/3i} = -\frac{1}{2} - \frac{\sqrt{3}}{2}i.$$

The representation in the complex plane of these three roots can be seen in figure 40.

Figure 40



As you can see, the three roots form an equilateral triangle. Moreover, the vertices of this triangle are on the circle with center (0,0) and radius 1.

29. We first express the complex number 1 in exponential form:

$$1 = e^{0i} = e^{2\pi ki}, \quad k = 0, 1, 2, 3, \dots$$

Therefore, we want to solve:

$$z^n = e^{2\pi ki}, \quad k = 0, 1, 2, 3, \dots$$

for $n = 2, 4, 5$ and 6 . To do this at the same time, we apply the n th root to both sides of the equality:

$$(z^n)^{\frac{1}{n}} = z = e^{\frac{2\pi ki}{n}} = e^{2\pi k/n i}, \quad k = 0, 1, 2, 3, \dots$$

To find the square roots, we substitute $n = 2$:

$$z = e^{2\pi k/2 i} = e^{\pi k i}, \quad k = 0, 1, 2, 3, \dots$$

and we consider $k = 0, 1$.

$$k = 0 : \quad z_1 = e^{0i} = \cos(0) + i \sin(0) = 1$$

$$k = 1 : \quad z_2 = e^{\pi i} = \cos(\pi) + i \sin(\pi) = -1$$

In the same way, to determine the fourth roots, we substitute $n = 4$:

$$z = e^{2\pi k/4 i} = e^{\pi/2 k i}, \quad k = 0, 1, 2, 3, \dots$$

and we consider $k = 0, 1, 2, 3$.

$$k = 0 : \quad z_1 = e^{0i} = 1$$

$$k = 1 : \quad z_2 = e^{\pi/2 i} = \cos\left(\frac{\pi}{2}\right) + i \sin\left(\frac{\pi}{2}\right) = 0 + i = i$$

$$k = 2 : \quad z_3 = e^{\pi i} = \cos(\pi) + i \sin(\pi) = -1$$

$$k = 3 : \quad z_4 = e^{3\pi/2 i} = \cos\left(\frac{3\pi}{2}\right) + i \sin\left(\frac{3\pi}{2}\right) = 0 - i = -i$$

Now we obtain the fifth roots. First we substitute $n = 5$:

$$z = e^{2\pi k/5 i} = e^{2\pi/5 k i}, \quad k = 0, 1, 2, 3, \dots$$

and we consider $k = 0, 1, 2, 3, 4$.

$$k = 0 : \quad z_1 = e^{0i} = 1$$

$$k = 1 : \quad z_2 = e^{2\pi/5 i} = \cos\left(\frac{2\pi}{5}\right) + i \sin\left(\frac{2\pi}{5}\right) = 0,309 + 0,951i$$

$$k = 2 : \quad z_3 = e^{4\pi/5 i} = \cos\left(\frac{4\pi}{5}\right) + i \sin\left(\frac{4\pi}{5}\right) = -0,809 + 0,588i$$

$$k = 3 : \quad z_4 = e^{6\pi/5 i} = \cos\left(\frac{6\pi}{5}\right) + i \sin\left(\frac{6\pi}{5}\right) = -0,809 - 0,588i = \bar{z}_3$$

$$k = 4 : \quad z_5 = e^{8\pi/5 i} = \cos\left(\frac{8\pi}{5}\right) + i \sin\left(\frac{8\pi}{5}\right) = 0,309 - 0,951i = \bar{z}_2$$

Finally, we obtain the sixth roots. First we substitute $n = 6$:

$$z = e^{2\pi k/6 i} = e^{\pi/3 k i}, \quad k = 0, 1, 2, 3, \dots$$

and we consider $k = 0, 1, 2, 3, 4, 5$.

$$k = 0 : \quad z_1 = e^{0i} = 1$$

$$k = 1 : \quad z_2 = e^{2\pi/6i} = \cos\left(\frac{2\pi}{6}\right) + i \sin\left(\frac{2\pi}{6}\right) = \frac{1}{2} + \frac{\sqrt{3}}{2}i$$

$$k = 2 : \quad z_3 = e^{4\pi/6i} = \cos\left(\frac{4\pi}{6}\right) + i \sin\left(\frac{4\pi}{6}\right) = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$$

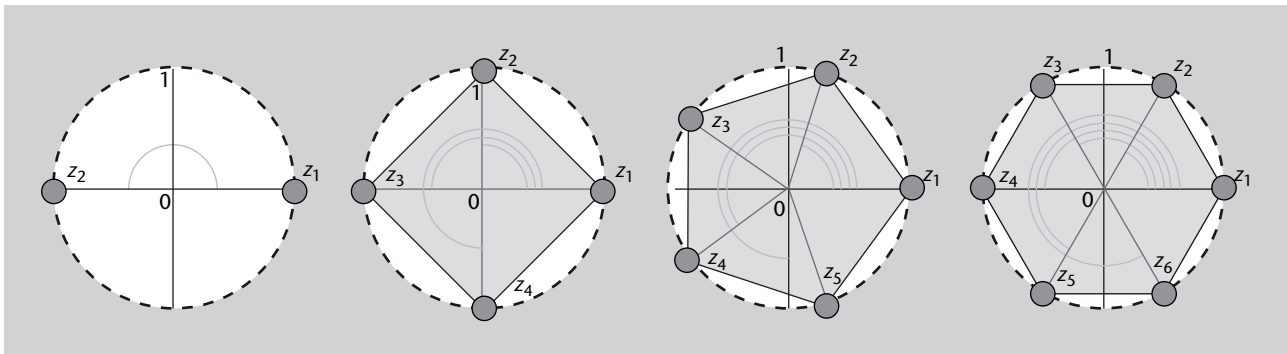
$$k = 3 : \quad z_4 = e^{6\pi/6i} = e^{\pi i} = -1$$

$$k = 4 : \quad z_5 = e^{8\pi/6i} = \cos\left(\frac{8\pi}{6}\right) + i \sin\left(\frac{8\pi}{6}\right) = -\frac{1}{2} - \frac{\sqrt{3}}{2}i = \bar{z}_3$$

$$k = 5 : \quad z_6 = e^{10\pi/6i} = \cos\left(\frac{10\pi}{6}\right) + i \sin\left(\frac{10\pi}{6}\right) = \frac{1}{2} - \frac{\sqrt{3}}{2}i = \bar{z}_2$$

In figure 41 we have plotted the square, fourth, fifth and sixth roots of the number $z = 1$. Notice that all the roots have the same modulus (they are located on a circle) and that they are located on the vertices of a regular polygon of n vertices. Thus, the square roots are always aligned, the fourth roots are the vertices of a square, the fifth roots are the vertices of a regular pentagon and the sixth roots are the vertices of a regular hexagon.

Figure 41



30. Now we will first solve the equation $z^4 + 16 = 0$. Note that this is equivalent to finding the values of z such that:

$$z^4 = -16$$

Therefore, we must calculate the fourth roots of the complex number -16 . This exercise can be done in a simple way using the following reasoning:

$$z^4 = -16 = 16(-1) \Rightarrow z = \sqrt[4]{16} \cdot \sqrt[4]{-1} = 2\sqrt[4]{-1}$$

Therefore, we can obtain the fourth roots of -16 from the fourth roots of -1 that we have already calculated before multiplying by two. In this way, the four fourth roots are:

$$z_1 = 2 \left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i \right) = \sqrt{2} + \sqrt{2}i$$

$$z_2 = 2 \left(-\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i \right) = -\sqrt{2} + \sqrt{2}i$$

$$z_3 = 2 \left(-\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i \right) = -\sqrt{2} - \sqrt{2}i$$

$$z_4 = 2 \left(\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i \right) = \sqrt{2} - \sqrt{2}i$$

Now, we can also do the same exercise step by step. So, the first thing we will do is to convert the number -16 to exponential form:

$$-16 = 16e^{\pi i} = 16e^{(\pi+2\pi k)i}, \quad k = 0, 1, 2, 3, \dots$$

Therefore, we want to solve:

$$z^4 = 16e^{(\pi+2\pi k)i}, \quad k = 0, 1, 2, 3, \dots$$

We now apply the fourth root to both sides of the equality:

$$\left(z^4\right)^{\frac{1}{4}} = z = (16e^{(\pi+2\pi k)i})^{\frac{1}{4}} = \sqrt[4]{16}e^{(\pi+2\pi k)/4i} = 2e^{(\pi/4+\pi k/2)i}, \quad k = 0, 1, 2, 3, \dots$$

To determine the fourth roots, we consider $k = 0, 1, 2, 3$.

$$k = 0: \quad z_1 = 2e^{\pi/4i} = 2 \cos\left(\frac{\pi}{4}\right) + 2 \sin\left(\frac{\pi}{4}\right)i = \sqrt{2} + \sqrt{2}i$$

$$k = 1: \quad z_2 = 2e^{(\pi/4+\pi/2)i} = 2e^{3\pi/4i} = 2 \cos\left(\frac{3\pi}{4}\right) + 2 \sin\left(\frac{3\pi}{4}\right)i = -\sqrt{2} + \sqrt{2}i$$

$$k = 2: \quad z_3 = 2e^{(\pi/4+\pi)i} = 2e^{5\pi/4i} = 2 \cos\left(\frac{5\pi}{4}\right) + 2 \sin\left(\frac{5\pi}{4}\right)i = -\sqrt{2} - \sqrt{2}i$$

$$k = 3: \quad z_4 = 2e^{(\pi/4+3\pi/2)i} = 2e^{7\pi/4i} = 2 \cos\left(\frac{7\pi}{4}\right) + 2 \sin\left(\frac{7\pi}{4}\right)i = \sqrt{2} - \sqrt{2}i$$

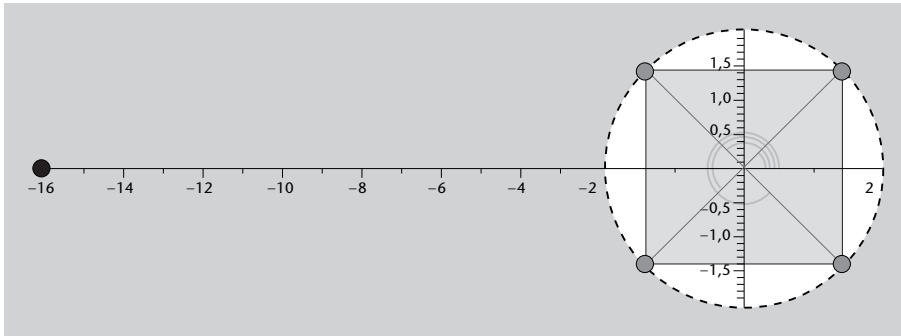
We see that we obtain the same results as with the simple way of performing the exercise.

Another simple way to calculate the fourth roots is considering that the four roots are the vertices of a square. Let us consider the number -16 , which has as module 16 and as angle π . We know that we can obtain the first root by taking the fourth root of the module and dividing by four the argument; in this way we have:

$$z_1 = \sqrt[4]{16}e^{\pi/4i} = \sqrt{2} + \sqrt{2}i$$

If we represent this point, we see that it is on the bisector of the first quadrant (the line that splits the first quadrant $y = x$). We plot now the circle of radius 2 and we place on this circle the four vertices of the square (you can see figure 42).

Figure 42



Graphically it can be seen that, if $z_1 = \sqrt{2} + \sqrt{2}i$, then to obtain z_2 we only have to change the sign of the x coordinate, that is, of the real part. In the same way, it is seen that $z_3 = -z_1$. Finally, to obtain z_4 what we must do is to change the sign of the y coordinate, that is, the sign of the imaginary part. In this way:

$$z_1 = \sqrt{2} + \sqrt{2}i$$

$$z_2 = -\sqrt{2} + \sqrt{2}i$$

$$z_3 = -z_1 = -\sqrt{2} - \sqrt{2}i$$

$$z_4 = \sqrt{2} - \sqrt{2}i$$

Now we will solve the equation $z^6 = i$. In this case, we are asked to find the sixth roots of the number i . The first thing to do is to convert this number to exponential form:

$$i = e^{\pi/2i}$$

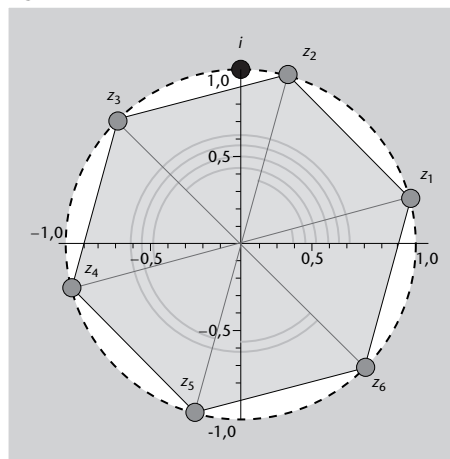
Therefore, since i has modulus 1, the sixth roots will have modulus $\sqrt[6]{1} = 1$. That is, the complex numbers we are looking for are on the circle of radius 1.

Remember that we obtain the first root by dividing the angle of the number by 6, that is:

$$z_1 = e^{\pi/2/6i} = e^{\pi/12i}$$

The other roots are found by placing them at the vertices of a regular hexagon, as seen in figure 43. That is, the difference in angle between each of the roots is $2\pi/6 = \pi/3$.

Figure 43



In this way:

$$z_1 = e^{\pi/12i}$$

$$z_2 = e^{(\pi/12+\pi/3)i} = e^{5\pi/12i}$$

$$z_3 = e^{(\pi/12+2\pi/3)i} = e^{3\pi/4i}$$

$$z_4 = e^{(\pi/12+3\pi/3)i} = e^{13\pi/12i}$$

$$z_5 = e^{(\pi/12+4\pi/3)i} = e^{17\pi/12i}$$

$$z_6 = e^{(\pi/12+5\pi/3)i} = e^{7\pi/4i}$$

Finally, we will find the solutions of the equation $z^3 + 8i = 0$. In this case, we must solve:

$$z^3 = -8i$$

that is, we must find the cube roots of $-8i = 8e^{-\pi/2i}$.

We know that the module of the three roots is $\sqrt[3]{8} = 2$; therefore, they are on the circle of center (0,0) and radius 2. We obtain the first root by dividing the initial angle by 3, that is:

$$z_1 = 2e^{-\pi/2/3i} = 2e^{-\pi/6i}$$

In this case, since the roots are on a triangle, the angle difference between the points is $2\pi/3$. This results in the following:

$$z_1 = 2e^{-\pi/6i}$$

$$z_2 = 2e^{(-\pi/6+2\pi/3)i} = e^{\pi/2i}$$

$$z_3 = 2e^{(-\pi/6+4\pi/3)i} = e^{7\pi/6i}$$

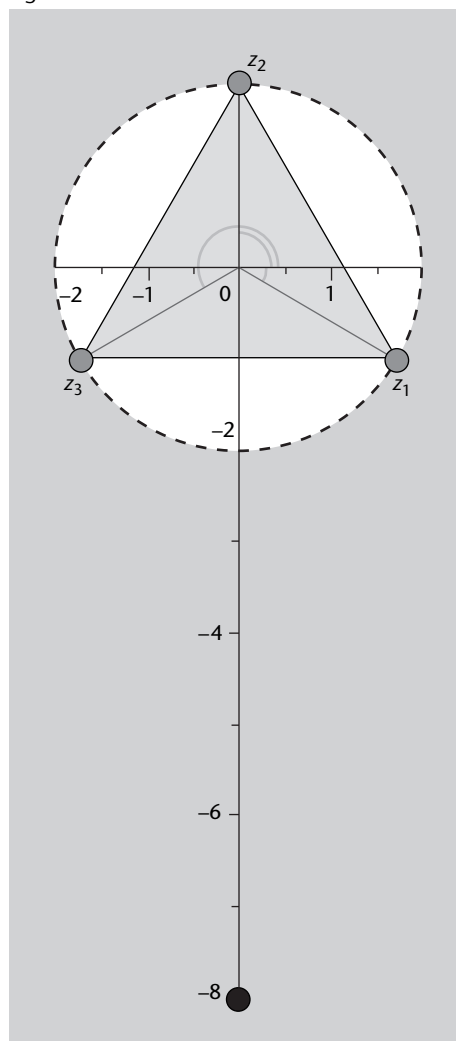
As we can see in figure 44, if we add z_1 to the angle of z_1 we obtain an angle of z_2 , that is, the second root is on the positive y -axis. This means that it is a pure imaginary number with positive imaginary part; therefore, $z_2 = 2i$. Furthermore, from figure 44 we see that to obtain z_3 we only need to change the sign of the real part of z_1 . Therefore:

$$z_1 = \sqrt{3} - i$$

$$z_2 = 2i$$

$$z_3 = -\sqrt{3} - i$$

Figure 44



Glossary

Argument of a complex number If we consider a complex number $z = a + bi$, angle formed by the positive x -axis and the vector (a, b) .

Complex number Number of the form $a + bi$, in which a and b are real numbers (called the *real part* and the *imaginary part*, respectively) and $i^2 = -1$. If $b = 0$, it is a real number and if $a = 0$, it is a pure imaginary number. The set of complex numbers is represented as $\mathbb{C}$.

Conjugate complex numbers A pair of complex numbers of the form $a + bi, a - bi$ with the same real part but with the imaginary part equal in magnitude but opposite in sign.

Imaginary part In a complex number $z = x + yi$, where x and y are two real numbers, the coefficient of the imaginary unit, also represented as $\text{Im}(z)$.

Imaginary unit Said particularly of the complex unit $i = \sqrt{-1}$.

Modulus of a complex number If we consider a complex number $z = a + bi$, the square root of $a^2 + b^2$. It is represented as $|z|$.

Principle of mathematical induction A mathematical proof technique useful for proving properties related to natural numbers.

Real part In a complex number $z = x + yi$, where x and y are two real numbers, the number x , also represented as $\text{Re}(z)$.

Bibliography

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